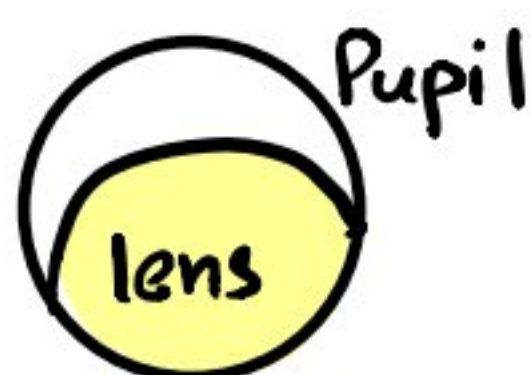


Acquired cause of Ectopia Lentis

- 1) Blunt trauma $\rightarrow$ posterior dislocation
- 2) Hypermature cataract $\rightarrow$ **MORGAGNIAN**
- 3) Ciliary body tumour
- 4) Pseudoexfoliation syndrome $\rightarrow$ Ab^(N) proteolytic material on zonules.

Symptoms

- i) ↓ sed vision
- ii) Unilateral diplopia



Mx :: ICCE

$\hookrightarrow$ The **only indication for ICCE is ectopia lentis**

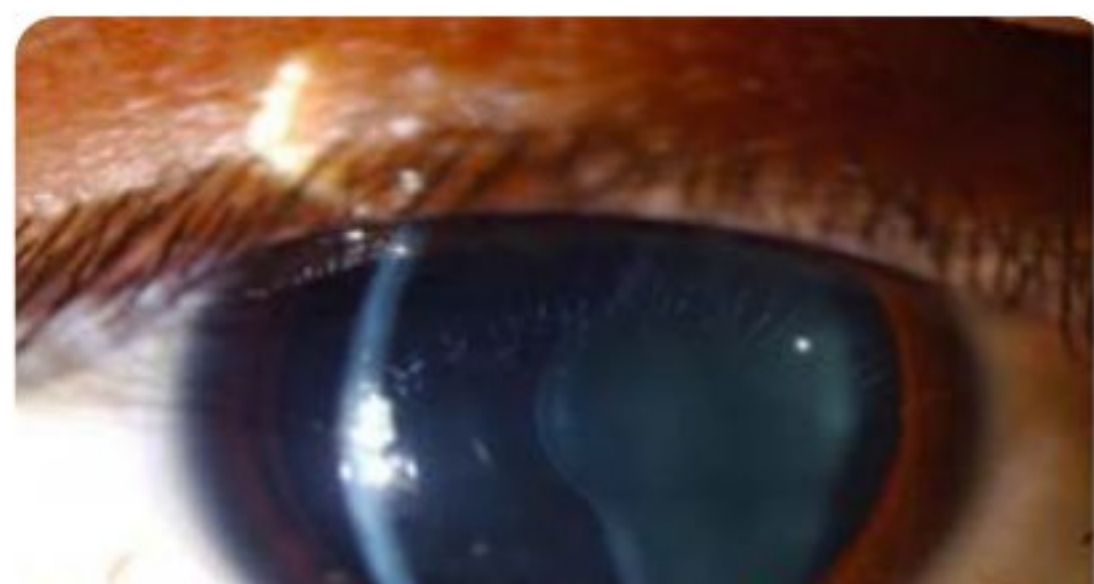
⊕ 2° Implantation $\rightarrow$ Scleral fixated PC-IOL or AC-IOL

LENTICONUS :: Conical projection of lens

- i) Anterior lenticonus
s/i Alport Sx



- ii) Posterior Lenticonus
s/i Lowe's Syndrome



Anterior lenticonus

Femtosecond laser $\rightarrow$ 1053 nm
 10^{-15} s

assisted phacoemulsification

- it can ::
- i) incision
 - ii) Ant. capsulorhexis
 - iii) Lens fragmentation

FLACS $\rightarrow$ Femtosecond laser assisted cataract surgery

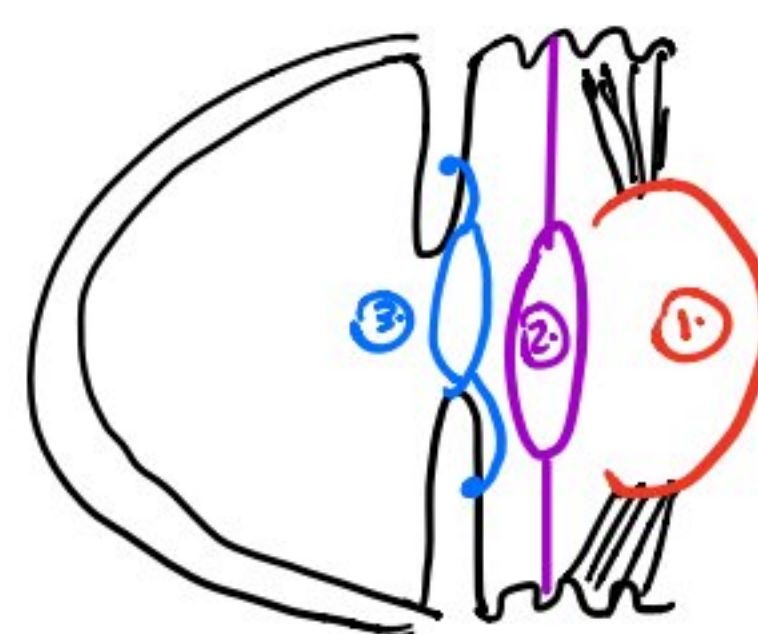
Best available procedure for cataract.

IOL Types of IOL ::

- i) PC-IOL { posterior chamber IOL }
- a) IOL behind the iris

i) **Capsular bag (ideal) = Power calculated by SRK formula**

- ii) Ciliary Sulcus
- iii) Iris-fixated
- iv) Scleral fixated IOL
{ when capsular bag absent }

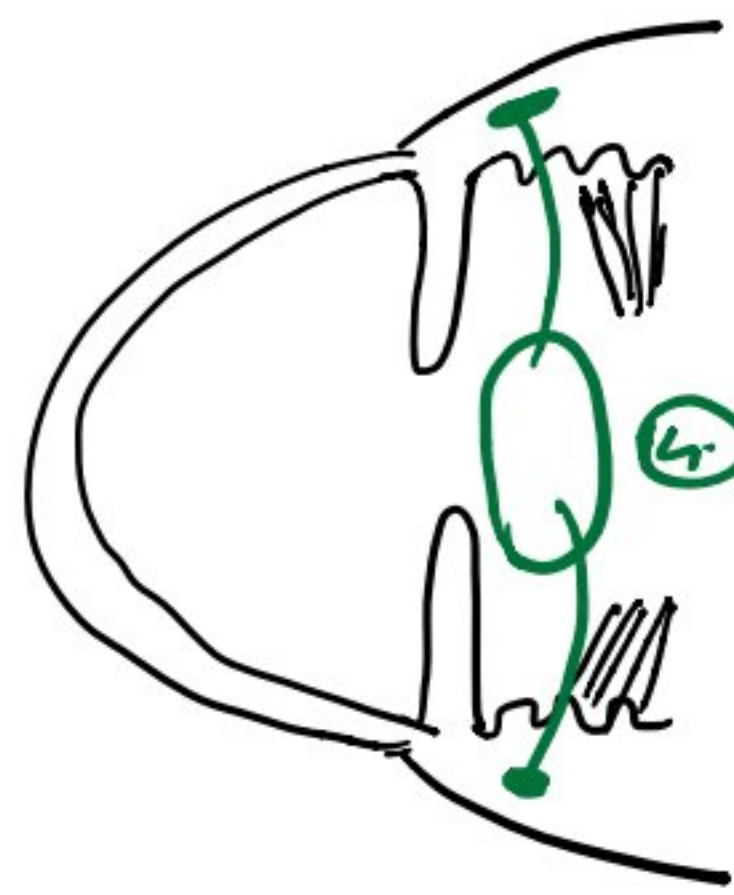
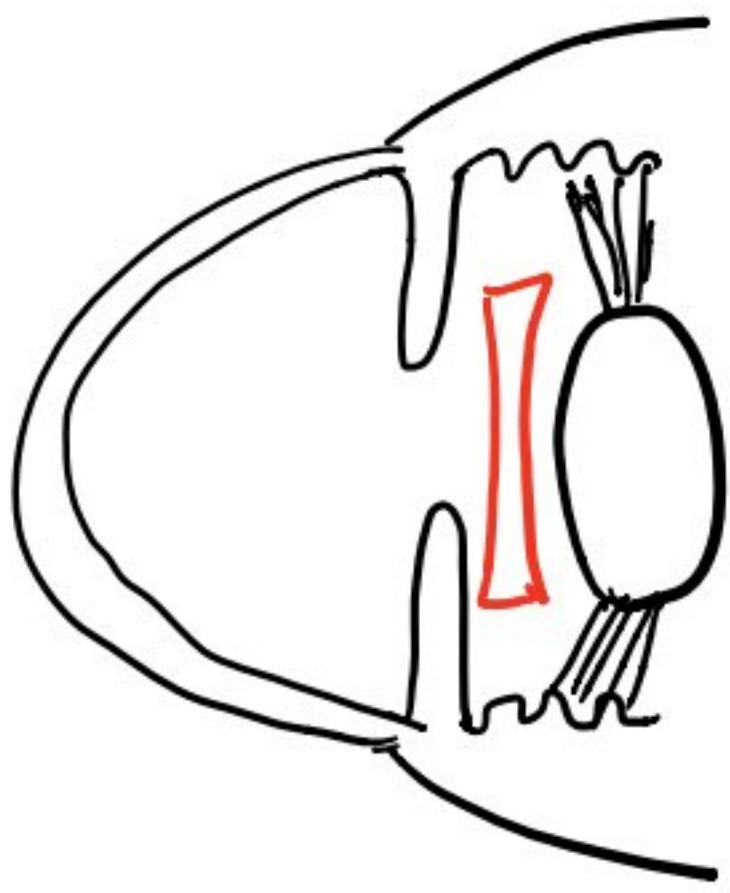


Recent NEET

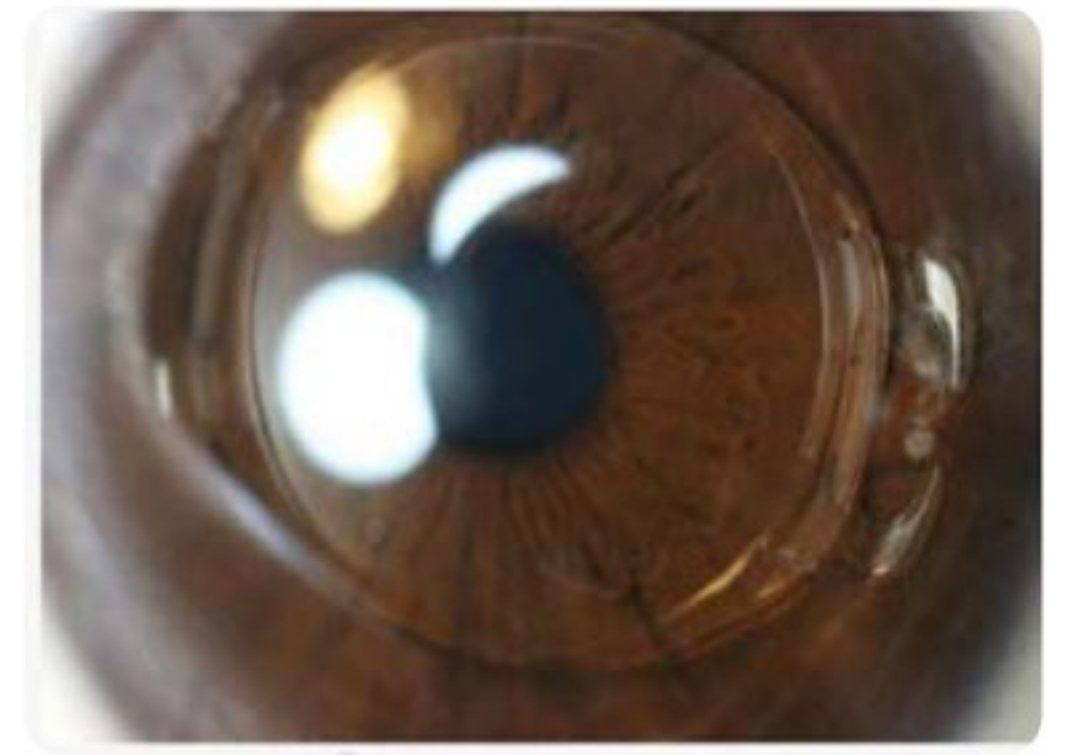


Mod. C loop PCIOL

b) Phakic IOL : Put over and above natural lens



Recent NEET



Iris claw lens

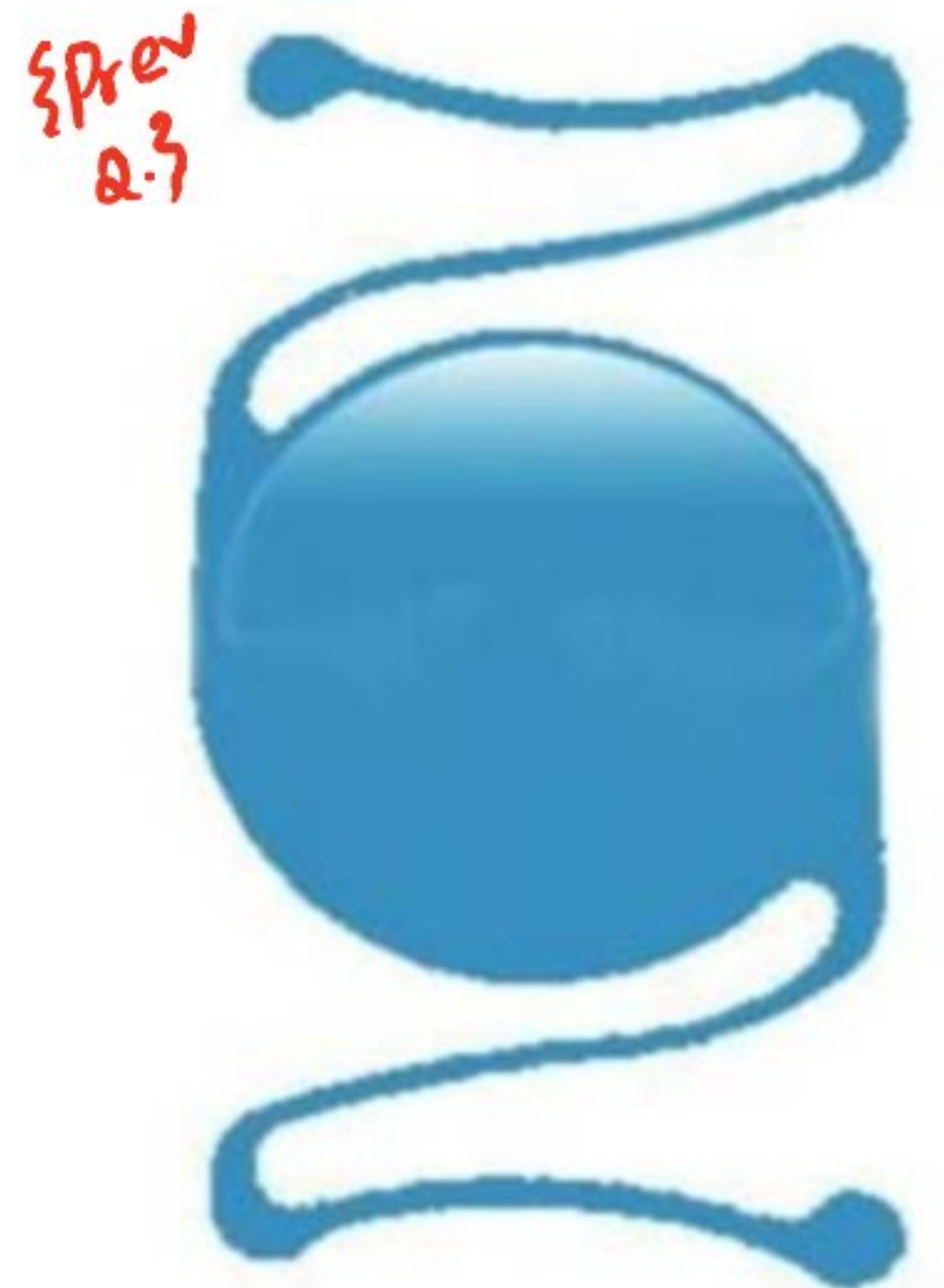
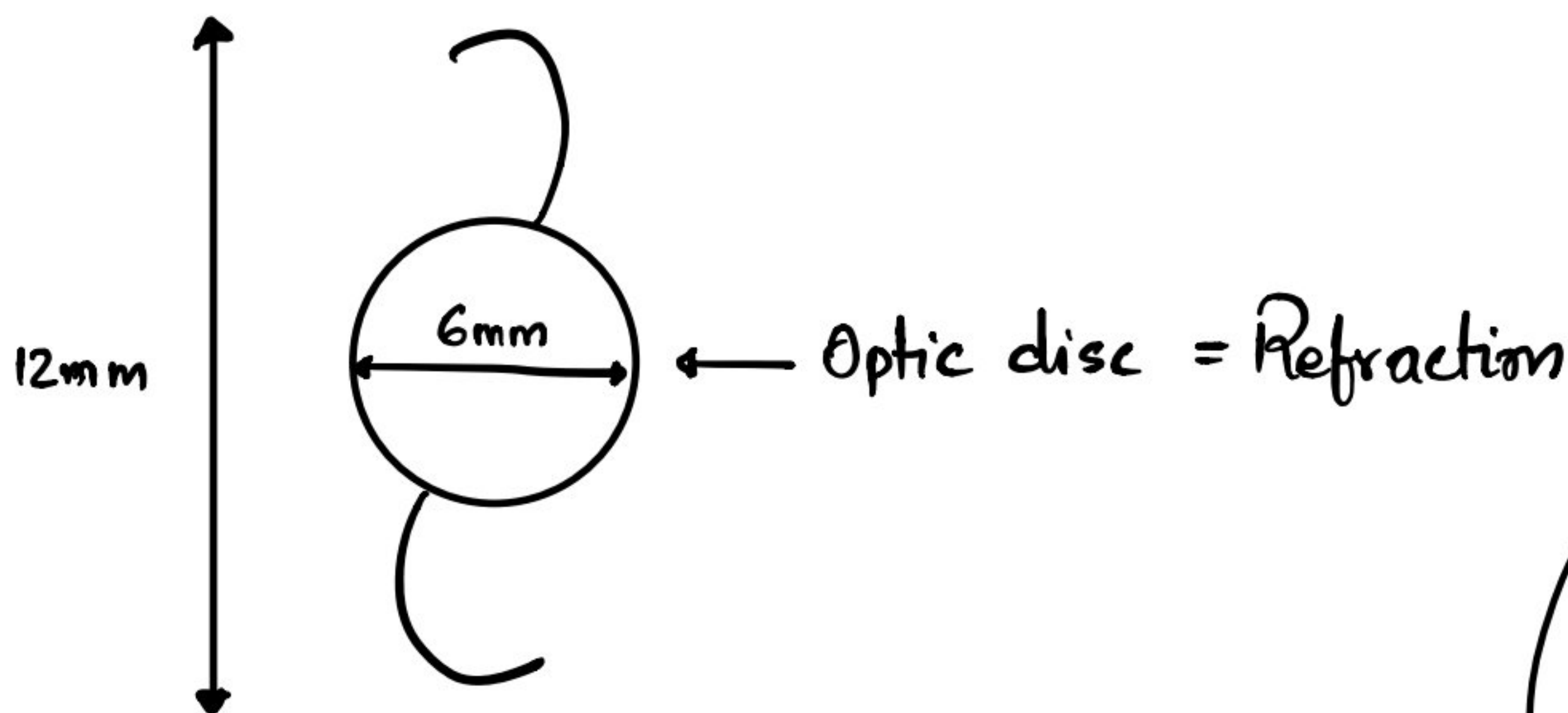
ii) AC-IOL : in front of Iris

- i) Angle supported
- ii) Iris supported
- iii) Iris fixated

Power = 3D lesser than PC-IOL

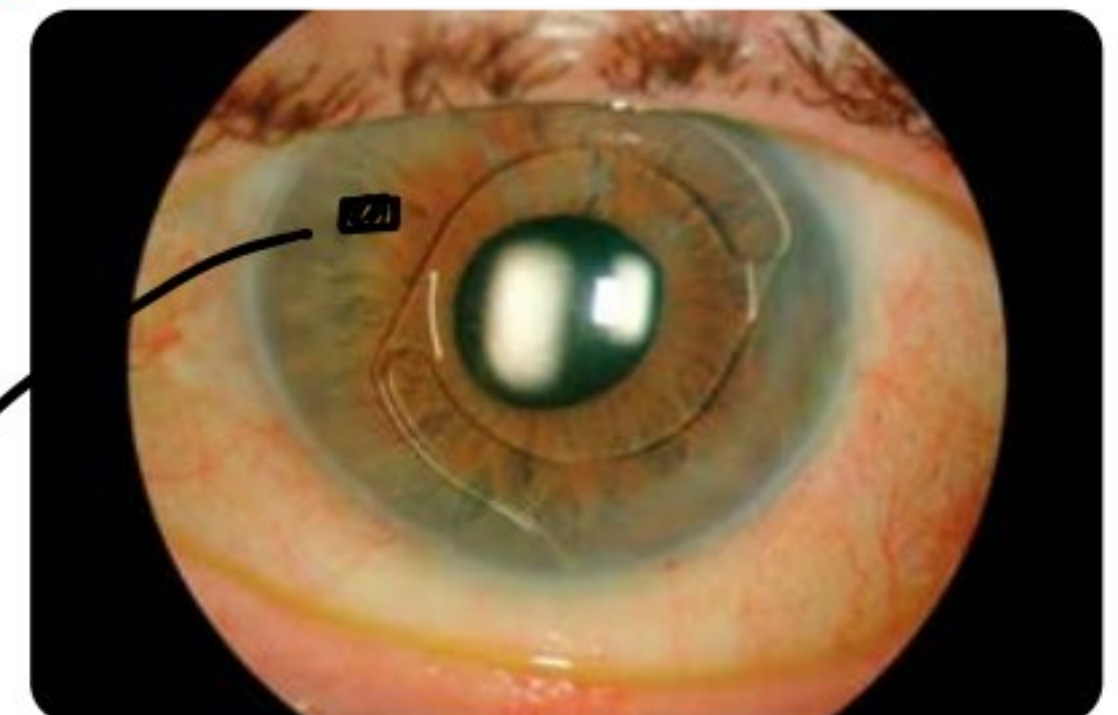
Complication of AC-IOL is UGH syndrome

Park of IOL :



Kelman multiflex AC IOL

§ Prev. Q3



AC-IOL

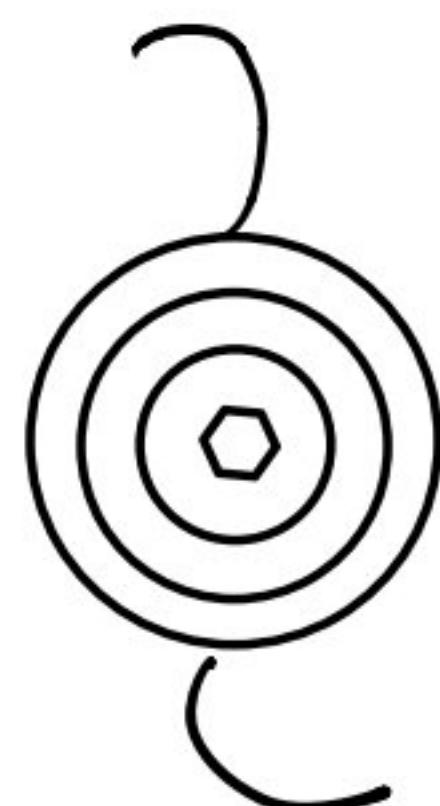
Peripheral Iridectomy
↳ for Aqueous to flow into AC

IOL based on optics :

- i) Monofocal IOL → . for distant vision
- still need glasses for N.V § after 6wk → after steroid dose?

ii) Multifocal IOL →

↓
Spectacle free



- i) Refractive IOL
- ii) Diffractive IOL
- iii) Accommodating IOL → dynamically change the accommodation by itself.

iii) Toric IOL: Correcting pre-existing astigmatism

BASED ON MATERIAL IOL

Rigid

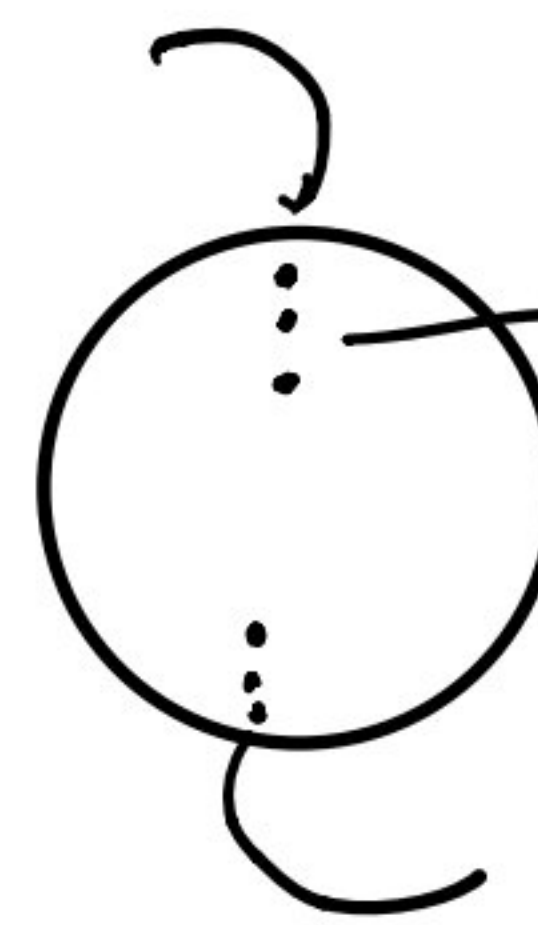
- PMMA

Foldable

- Incision size reduced
- Acrylic
- Silicone
- Hydrogel

Rotable

- Least size
- Acrylic



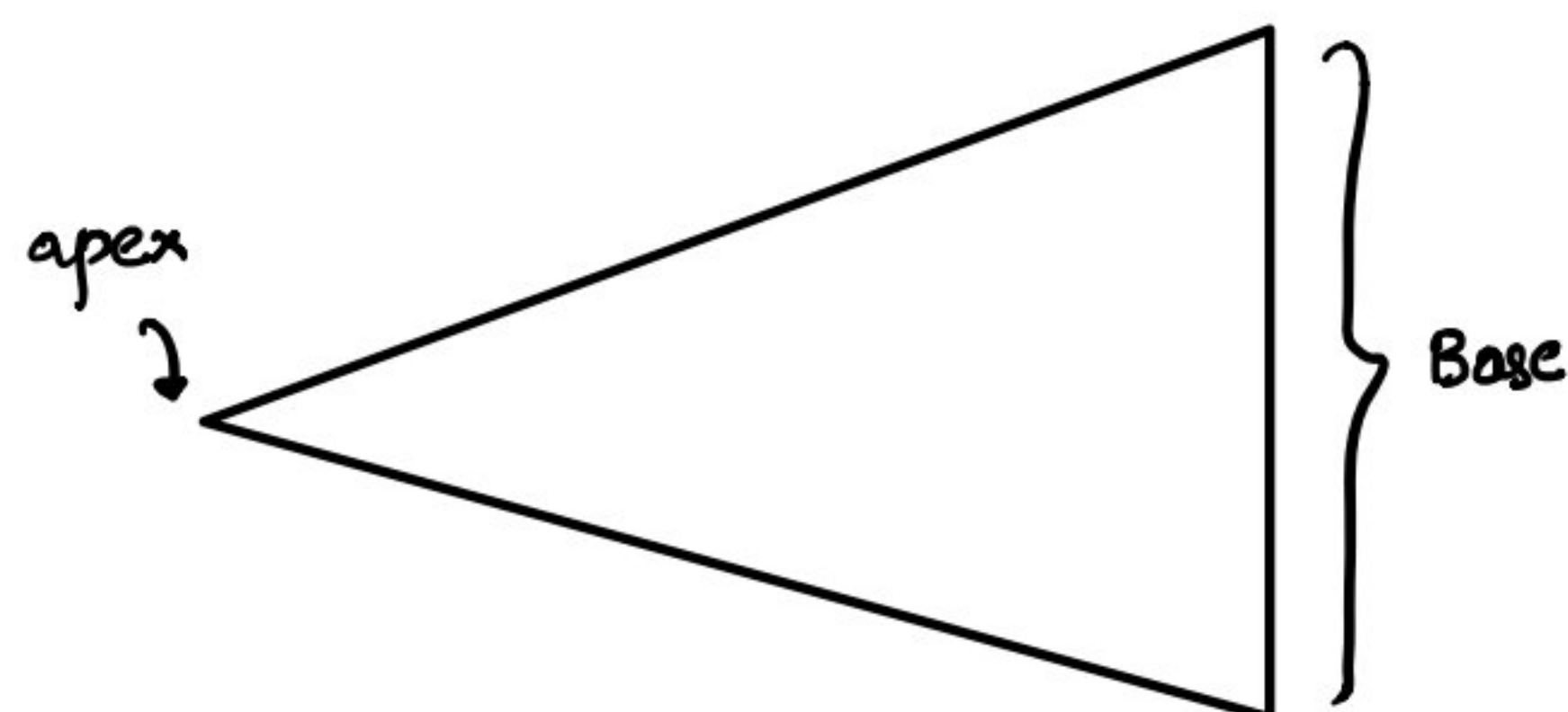
dotted line indicate the axis of Toric IOL and it is aligned to coincide with the steep axis of patient eye

Best material for IOL
↳ **HYDROPHOBIC ACRYLIC**

Orbit:

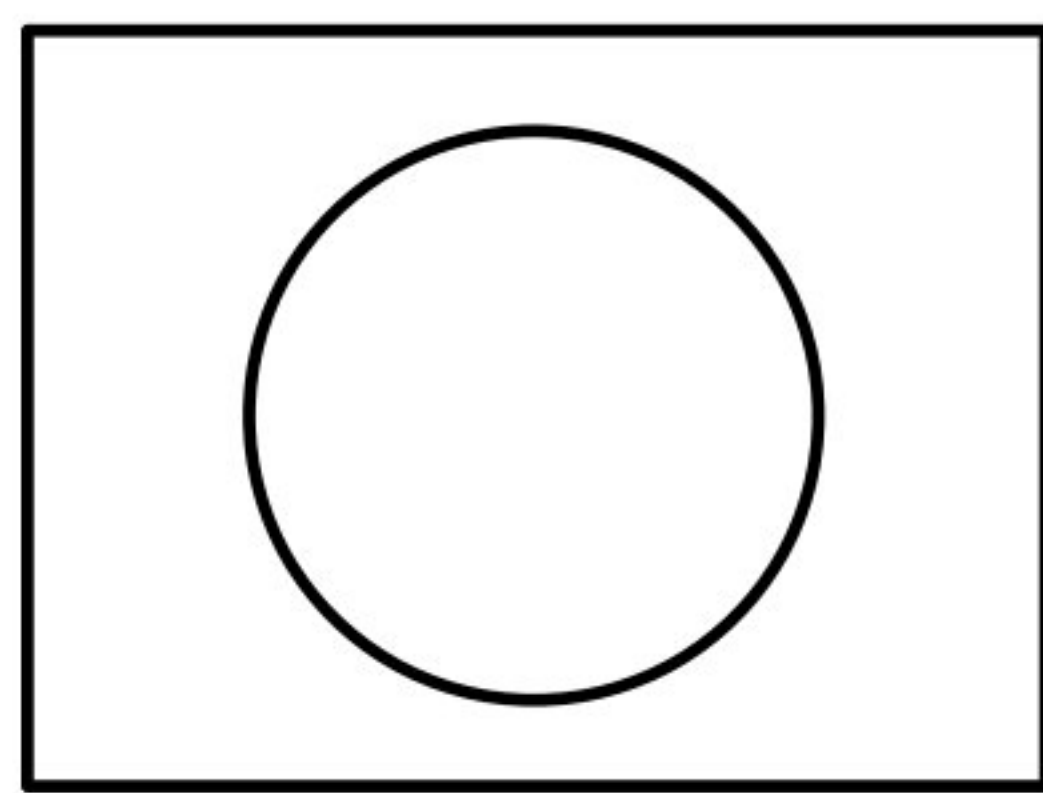
Shape: pyramid

volume: 30ml



Lesser wing of sphenoid
Frontal

Zygomatic
Greater wing of
sphenoid.



PA → palatine
ZA → Zygomatic
MA → maxillary

Body of sphenoid

medial

L

lacrimal fossa
↳ holding lacrimal sac.

Ethmoid

lacrimal



ORBITAL CAVITY

A pyramidal space with a base, apex and four walls.

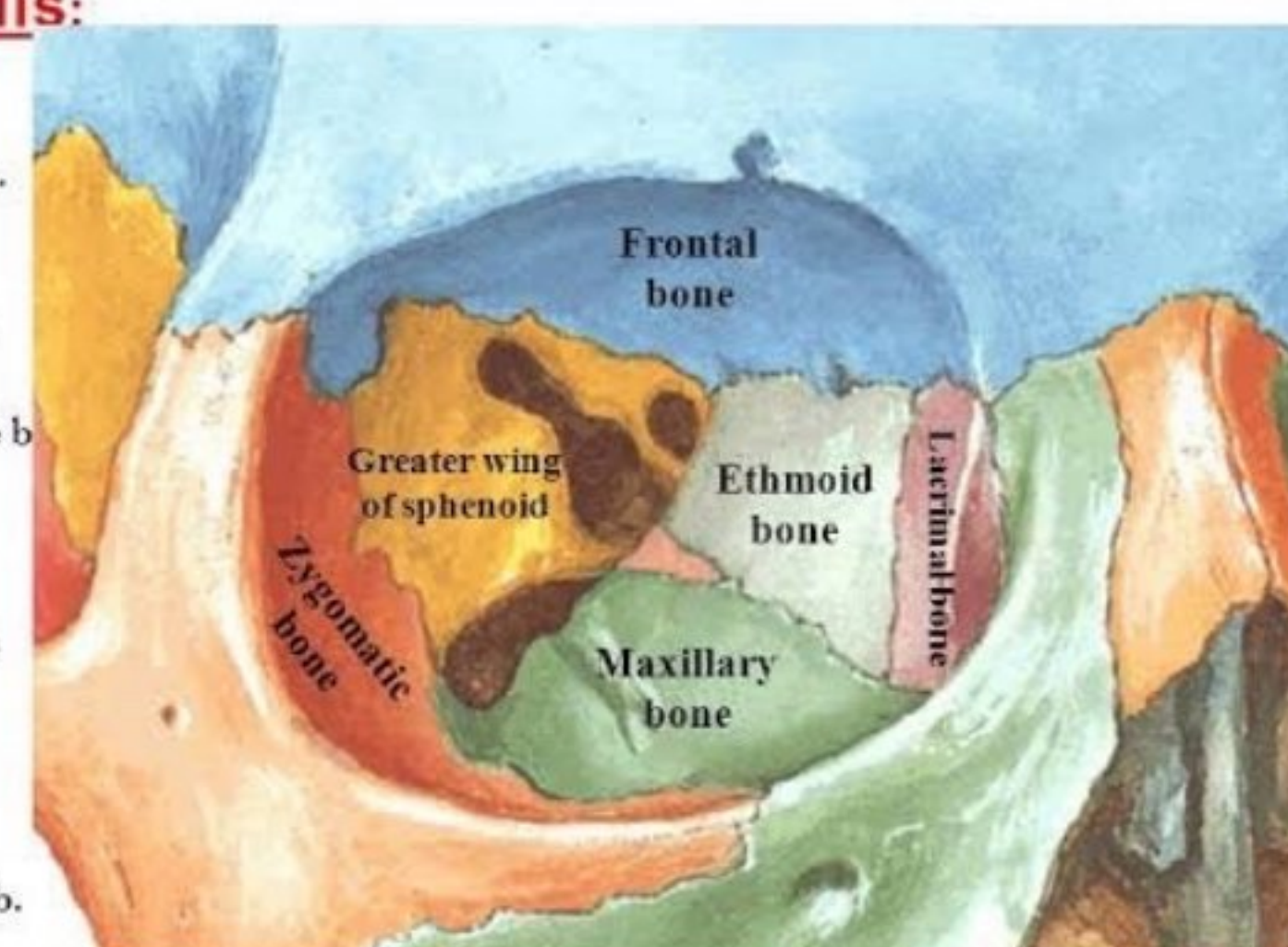
- 7 bone are present in the orbit
- medial wall thinnest → Lamina papyracea (thinnest bone ⇒ ethmoidal bone)

Children: Ethmoidal sinusitis → Mcc of orbital cellulitis in children.

g: weakest wall / (MC) wall to # → Floor of orbit d/t infraorbital groove

Boundaries and Walls:

- **Roof**
 1. Orbital plate of frontal b.
 2. Lesser wing of sphenoid b.
- **Floor**
 1. Orbital surface of maxilla
 2. Zygomatic bone
 3. Orbital process of palatine b.
- **Medial wall**
 1. Lacrimal bone
 2. Orbital plate of ethmoid b.
 3. Body of sphenoid b.
- **Lateral wall**
 1. Zygomatic b.
 2. Greater wing of sphenoid b.



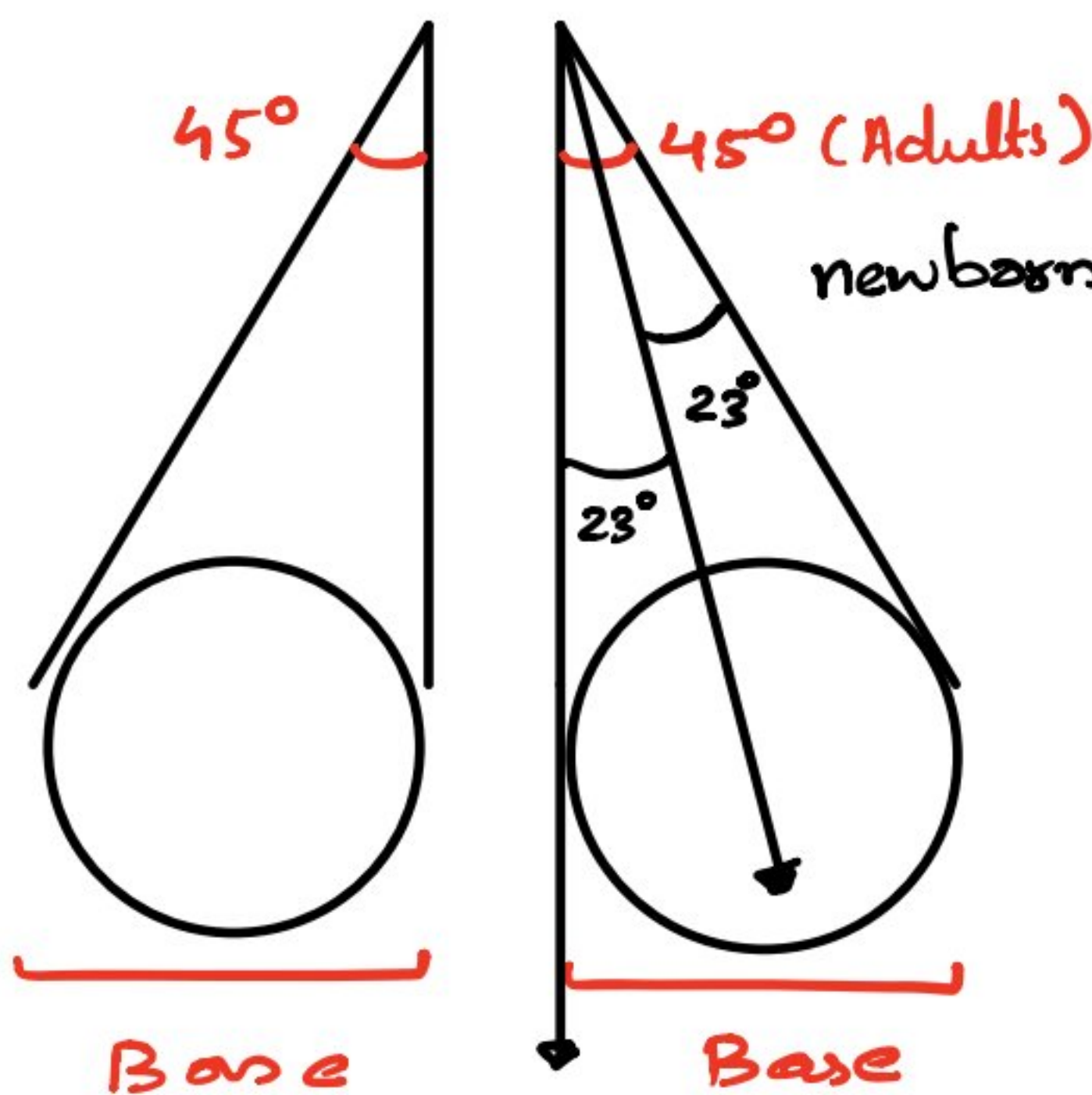
- Lateral wall strongest, thickest.
- weakest point of orbit → posteromedial aspect of maxillary bone in floor
- Blunt trauma → Floor #
or
Blow out #

, Hypo-esthesia or anaesthesia of ipsilateral cheek

Xray sign ⇒ Tear drop sign
(waters view)

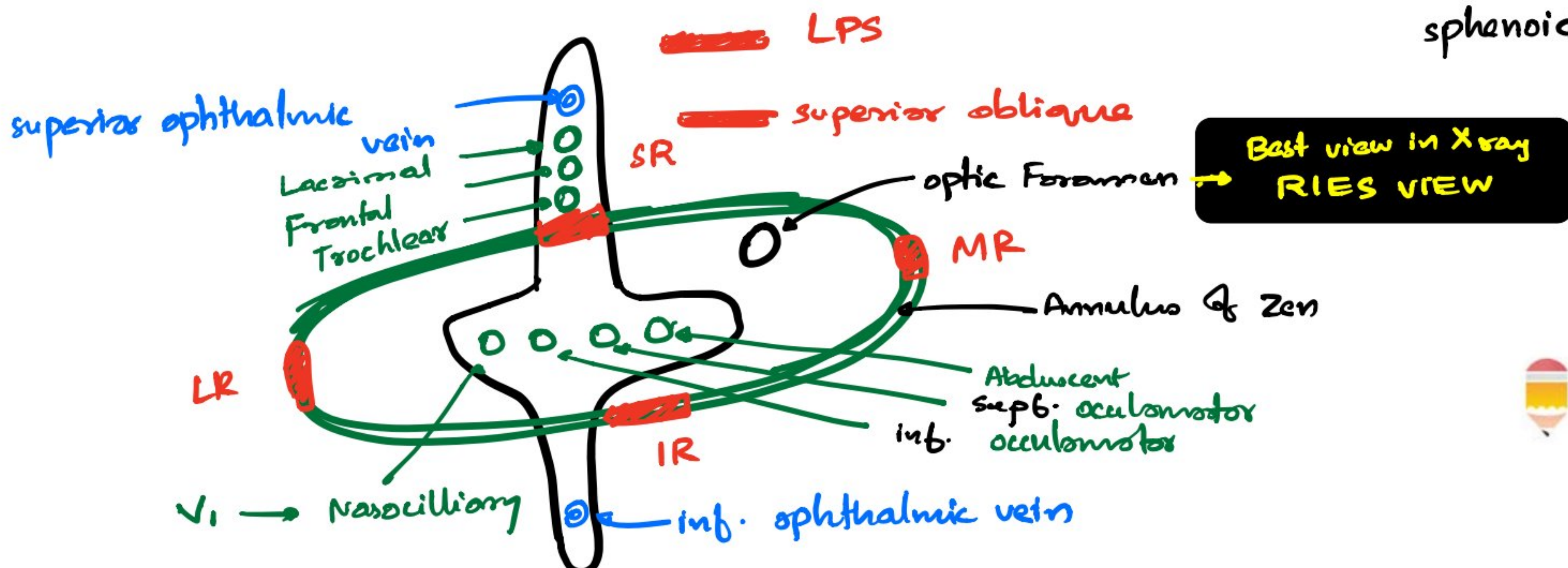
• enophthalmos

• inferior rectus can get entrapped in this # } Restrictive Squint
↳ difficulty to look up



Eyeball & orbit are at 23°

- 3 gap b/w orbit and brain → 1) Optic canal & optic N. & ophthalmic artery? Gap of lesser wing of sphenoid
- 2) Superior orbital fissure gap b/w lesser and greater wing of sphenoid
- 3) Inferior orbital fissure → on the floor b/w maxillary & greater wing of sphenoid



Superior Orbital fissure Syndrome:

• U/L condition

C/F: Pushes the eyeball → proptosis
3, 4, 6 ophthalmoplegia
loss of sensation of cornea → V1

Intra-orbital fissure ⇒ @ floor of orbit
→ contains the infra orbital N.

↓
Branch of Maxillary division of Vth N. (V2)
→ sensation to cheek

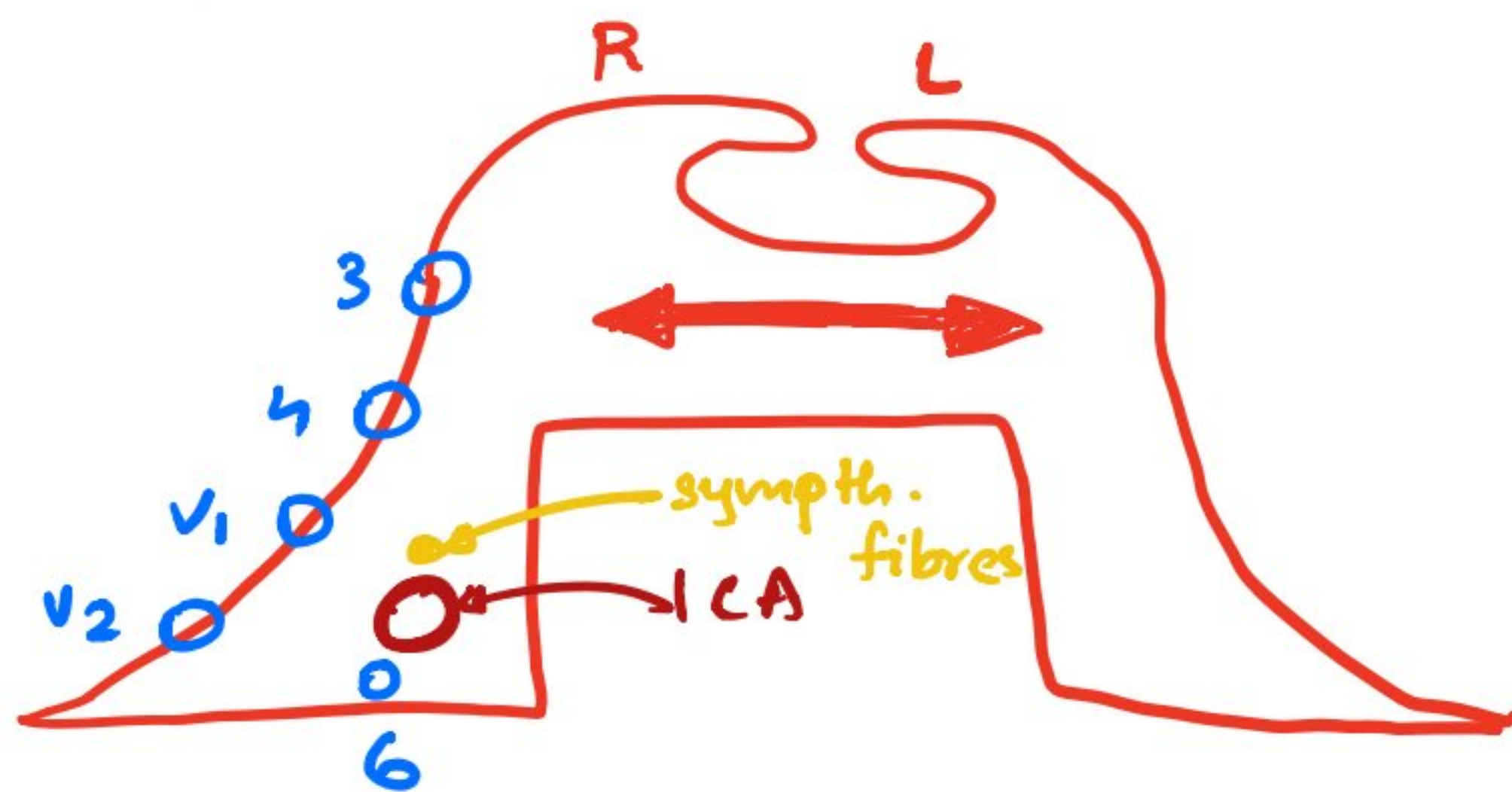
Orbital Apex Syndrome: U/L condition

CF: Pushes the eyeball → Proptosis

- 3, 4, 6 ophthalmoplegia
 - Loss of sensation of cornea → V₁
- ⊕ Blindness & Optic N. 3

Cavernous Sinus syndrome:

- Any infection / tumour in Cavernous Sinus



CF: 1st N. affected → V₁ N
1st Muscle " → LR
1st eye movement " → Abduction

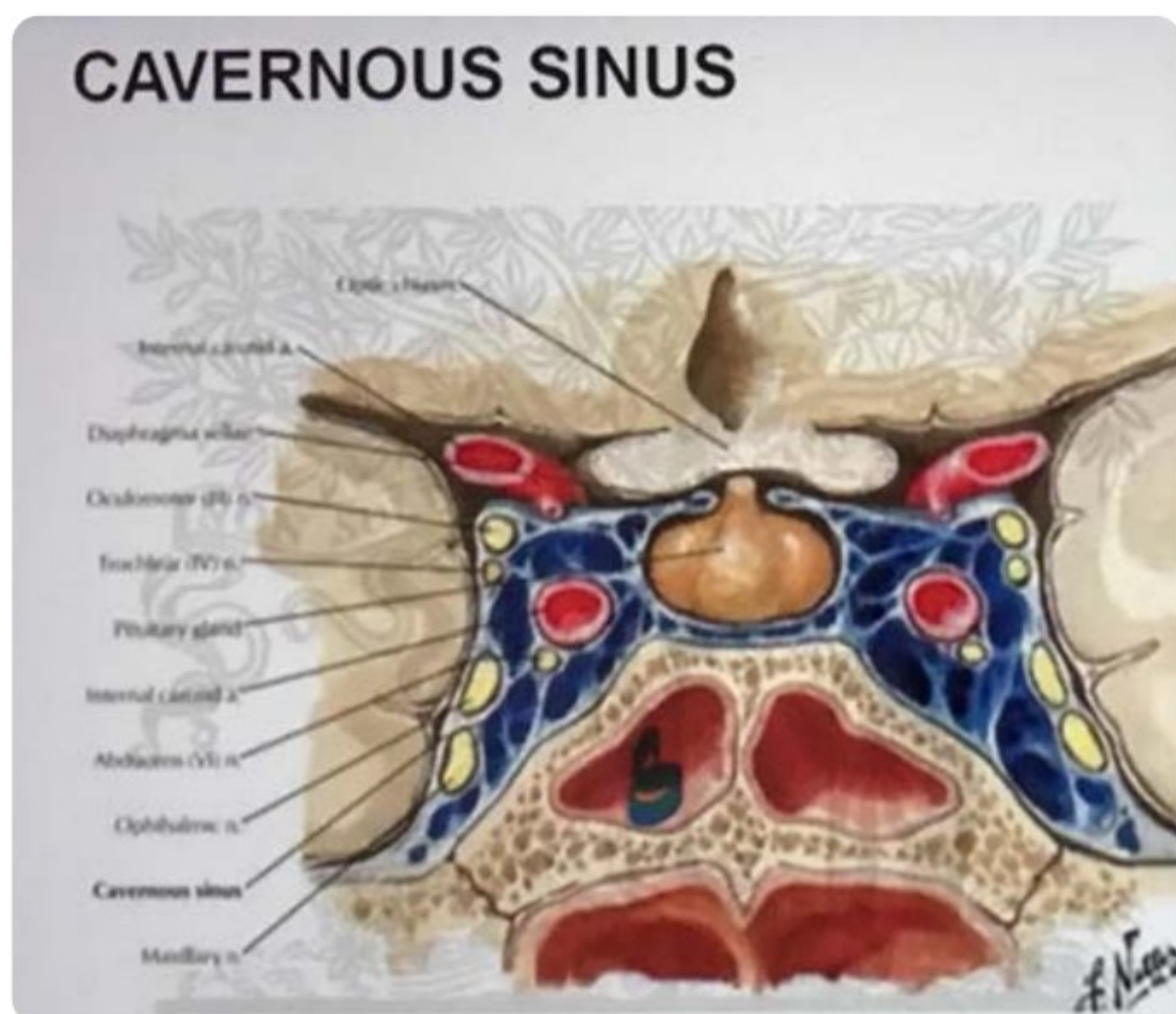
CF: i) Proptosis
ii) Ophthalmoplegia
iii) Loss of sensation of cornea / cheeks +/-
Long ciliary branch of Nasociliary N. (V₁)
Infraorbital N. (V₂)
iv) Horner's Sx & when sympathetic fibres affected
↓
Postganglionic Type 3

Cavernous sinus Thrombosis:

HOCKEY STICK SIGN → Cavernous Sinus Thrombosis (MR Angio)
d/t s.d.v dilatation

* Hockey stick sign s/i CRETZ FIELD JACOB'S D/s.

- Pathognomonic sign of CST → **MASTOID TENDERNESS** d/t Mastoid emissary veins.
- CST (B/L) → painful B/L proptosis
↳ (MCC) of B/L orbital cellulitis



Cavernous Sinus thrombosis

Q: What is the first sign of B/L CST?

a) Proptosis

b) V/L abduction deficit

c) C/L adduction deficit

d) C/L abduction deficit → 1st sign after spread

It only said first sign of CST

AIMS

TOLOSA HUNT SYNDROME → Non specific granulomatous inflammation affecting the cavernous sinus.

Presentation:

→ • U/L headache
• multiple cranial palsies
• external ophthalmoplegia
• Proptosis } cavernous sinus Sx.

Rx: Systemic steroids

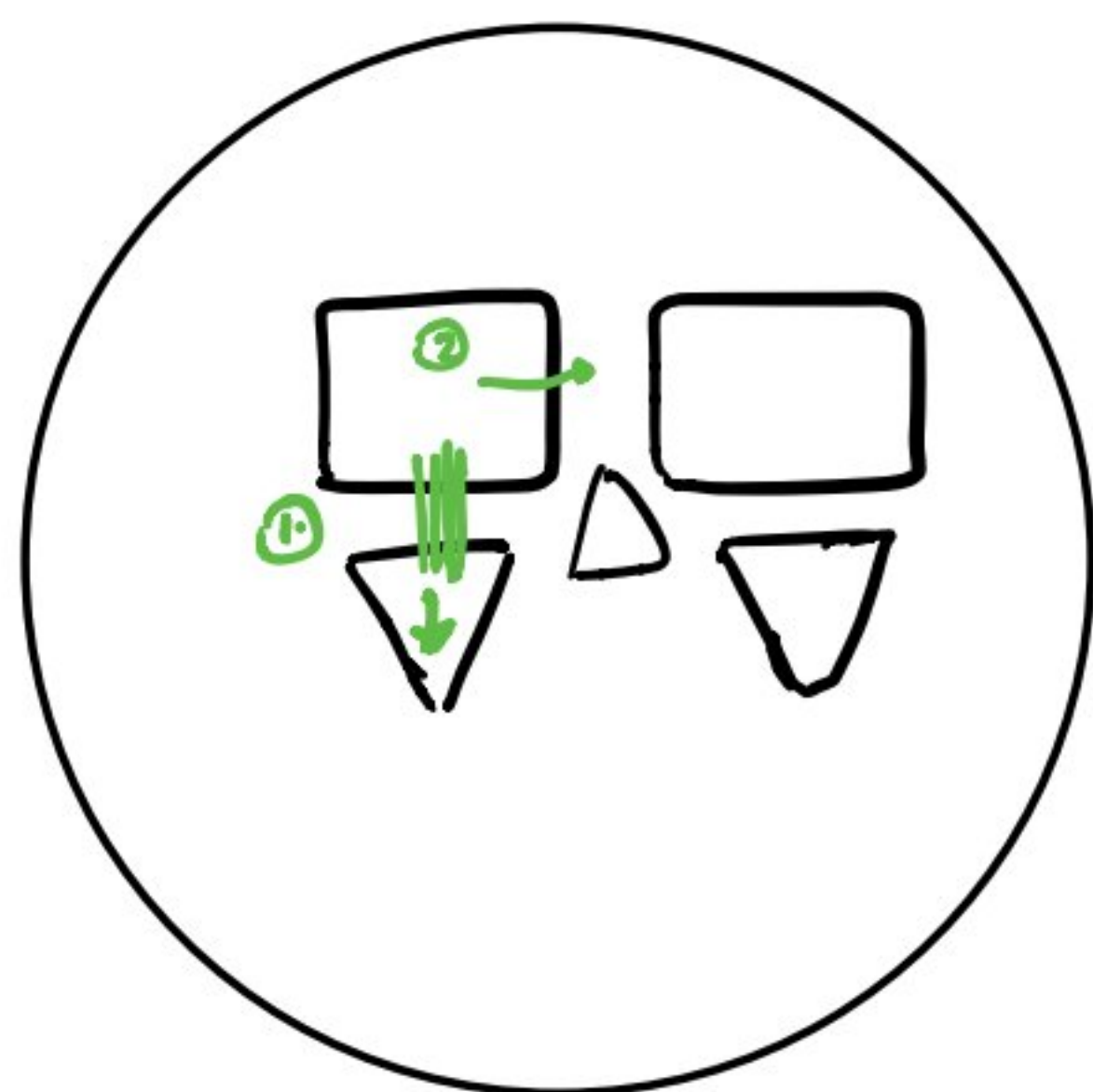
BLOW OUT #: Orbital content prolapse out of the orbital cavity

Pure

Orbital rim is intact.

Impure

a/w orbital rim #



Inferior orbital fissure
gap b/w lateral wall & floor.

Passes → • Infraorbital vessels / nerve → sensation of cheek.
• Zygomatic nerve
• pterygopalatine ganglion

C/F:

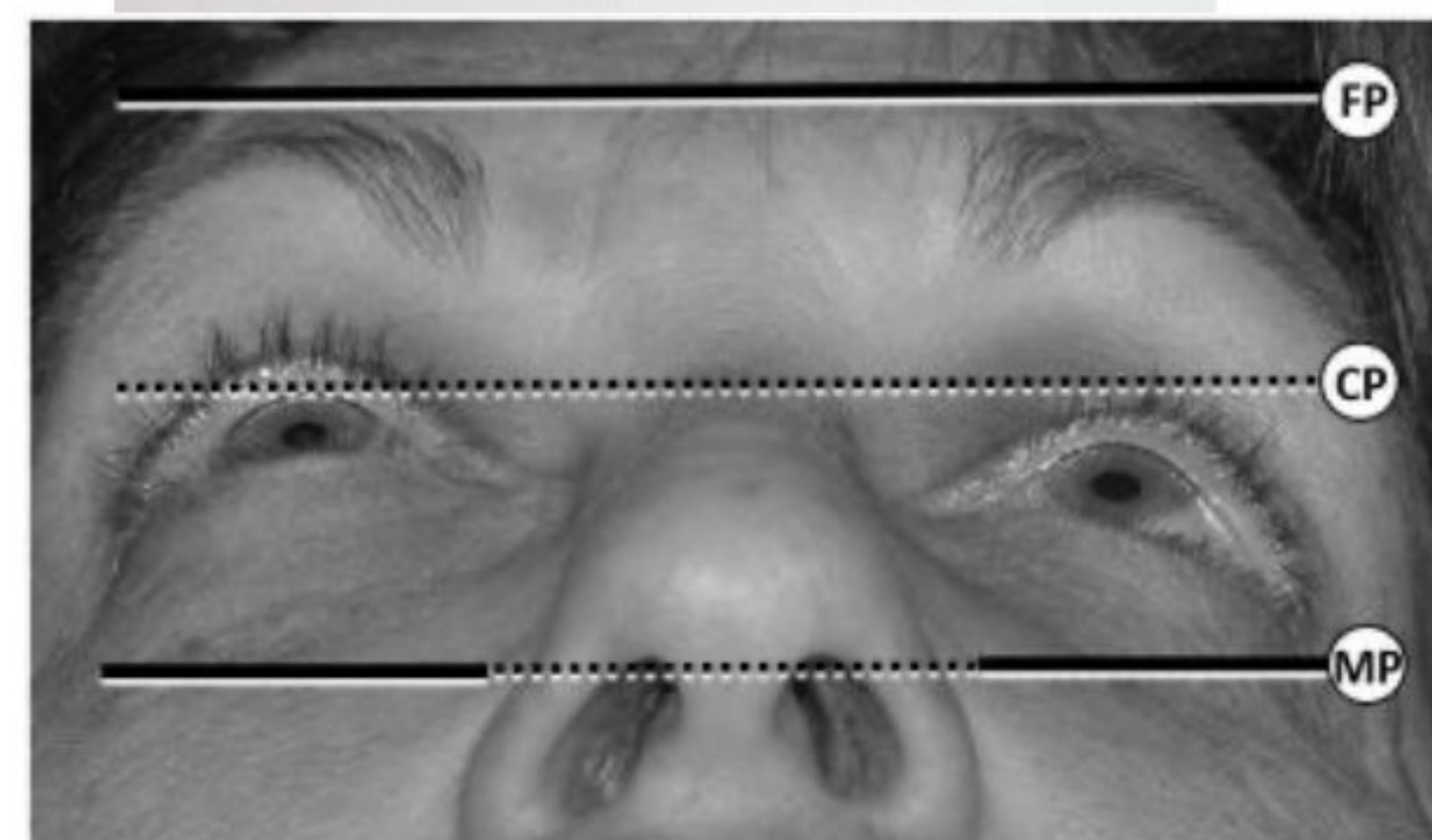
• Hypo / Anaesthesia of cheek & d/t infraorbital N. involvement } characteristic sign

InxOC: PLAIN CT of orbit



↓ se in Orbital cellulitis
ENOPHTHALMOS

ENOPHTHALMOS



Worms view



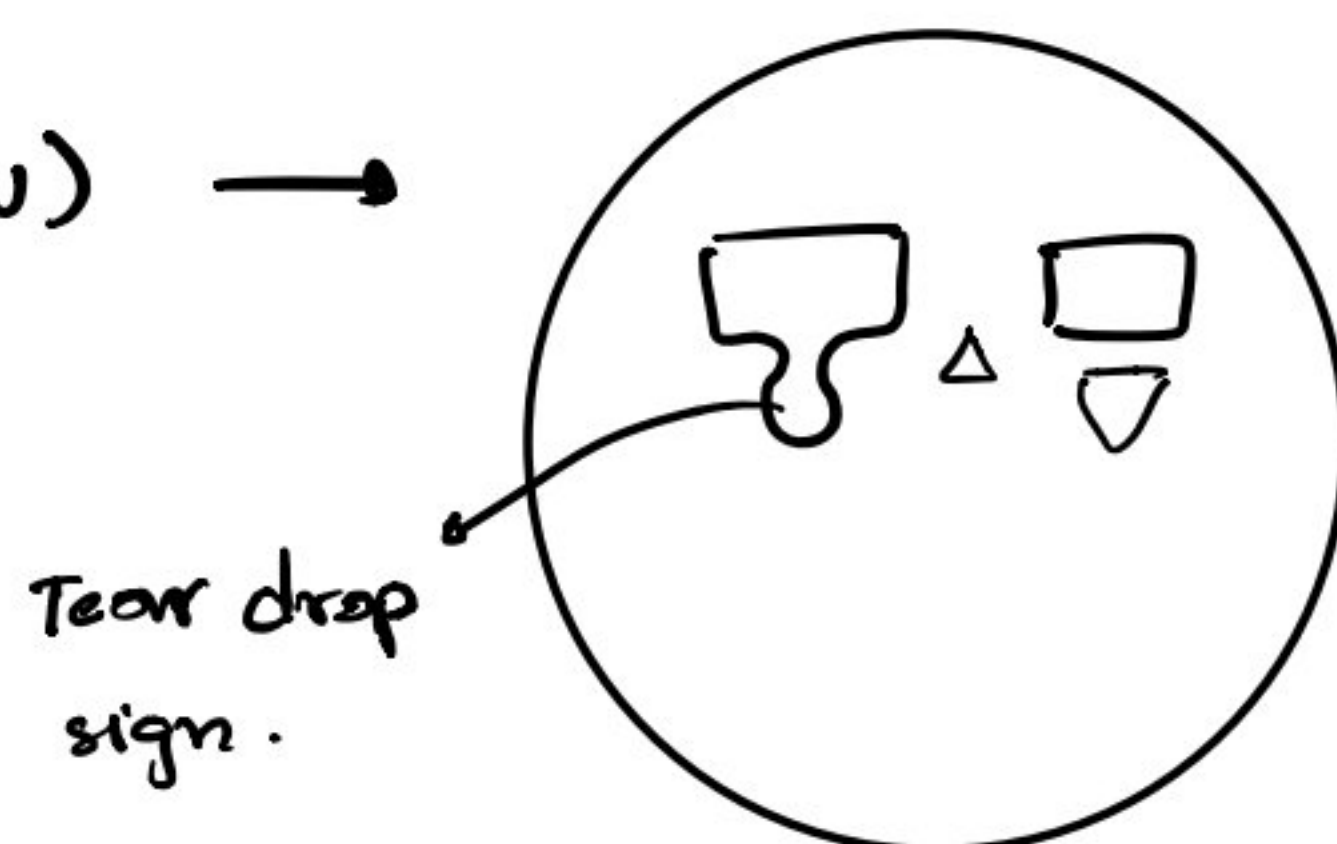
Normal CT

• Entrapment of IR in # site → Restriction of upgaze



DIPLOPLIA (vertical)
Restrictive Squint

Xray (water's view) →



Tear drop sign.

Management : **Expectant Mx**

- Observation
- Systemic steroids → short course → to address orbital edema

Active Mx / floor Repair req → with persistence of symptoms

indications

- 1) Enophthalmos > 2mm
- 2) Persistent diplopia > 2wk

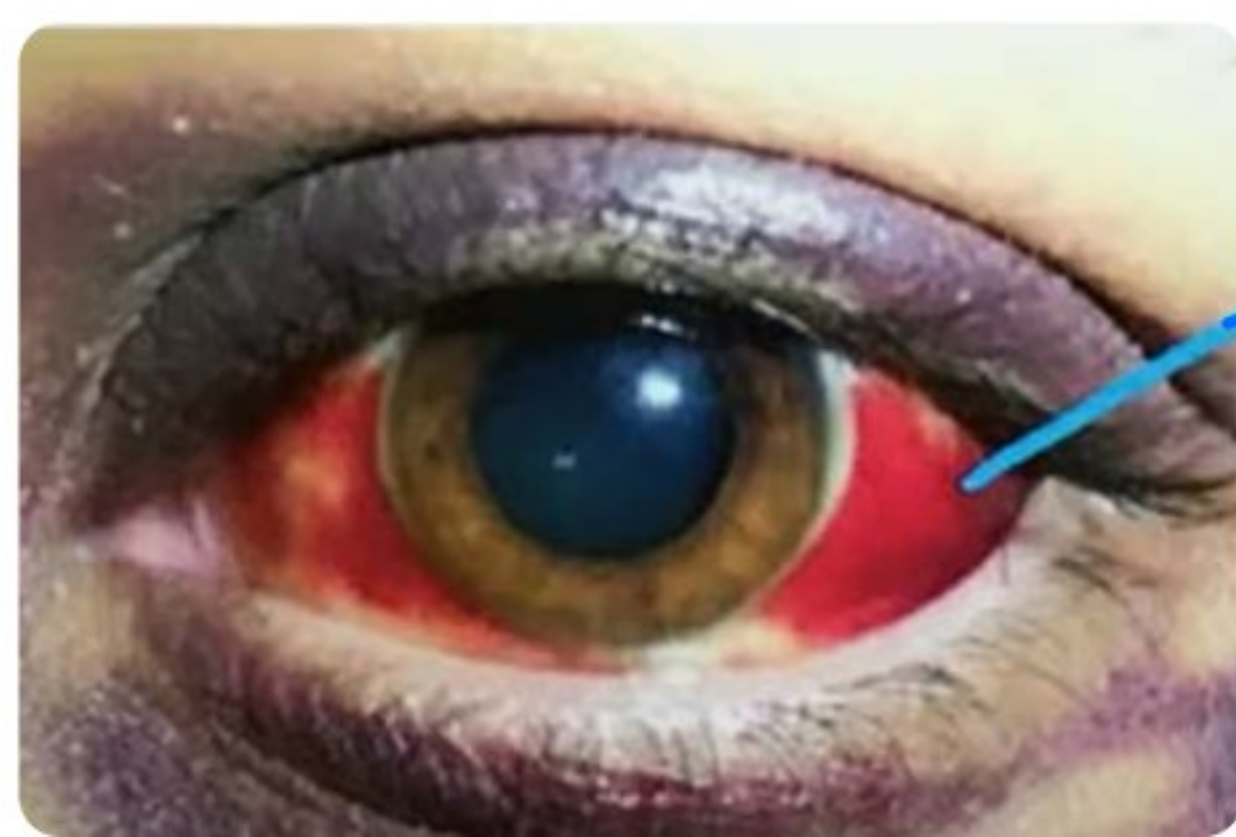
EYE FINDINGS OF BLUNT TRAUMA

1) Periorbital Ecchymosis



→ Raccoon's Eye
/ Panda Eye

- 2) Chemosis → Edema of conjunctiva
- 3) Subconjunctival H'mage



Bright Red: posterior extent is not visible
Basal skull #

Q: The most common site of globe rupture following trauma is ____.

- a) **Superonasal limbus**
- b) Inferonasal limbus
- c) Superotemporal limbus
- d) Inferotemporal limbus

• superonasal limbus is the **(mc)** site of globe rupture

This occurs d/t counter coup effect
↳ lower temporal quadrant is the most exposed to trauma.

4) Hyphaema → Major Arterial circle of Iris @ Iris root rupture

↓ f/b

Corneal Blood staining (permanent)
@ stroma level.

5) Iridodialysis → D shaped pupil
Detachment of Iris root for CB attachment

6) Traumatic tears of sphincter muscle → Mydriatic Pupil



Traumatic tears of sphincter muscle

TRAUMATIC MYDRIASIS

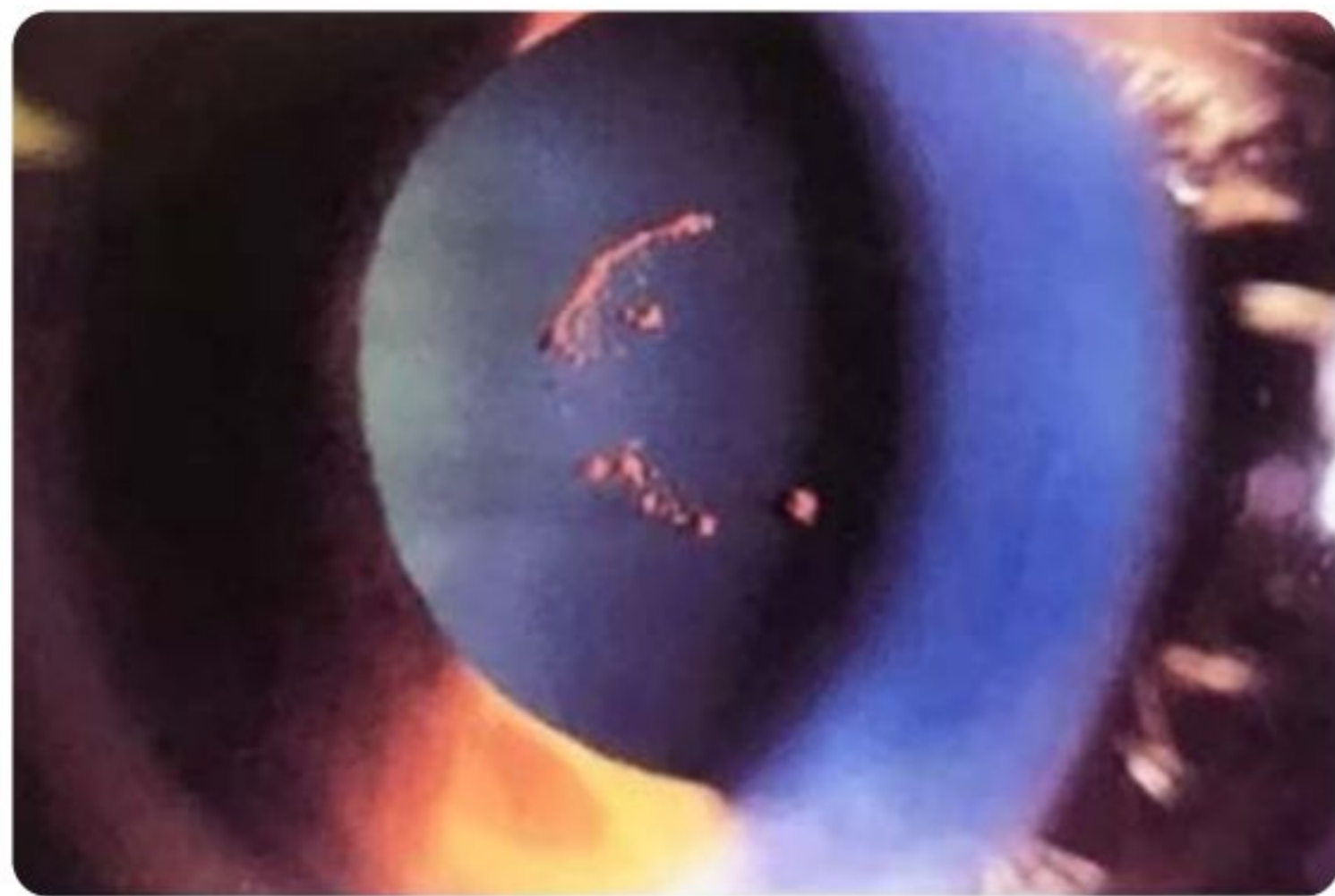
7) Angle recession on Gonioscopy → separation of circular & longitudinal fibres
of ciliary muscle on Gonioscopy

↓ e/d

↓
WIDENING OF C. Body band

Angle recession Glaucoma

8) Vossius Ring ← Blunt traum^e



↑↑ IOP

↓

• Sphincter spasm

↓

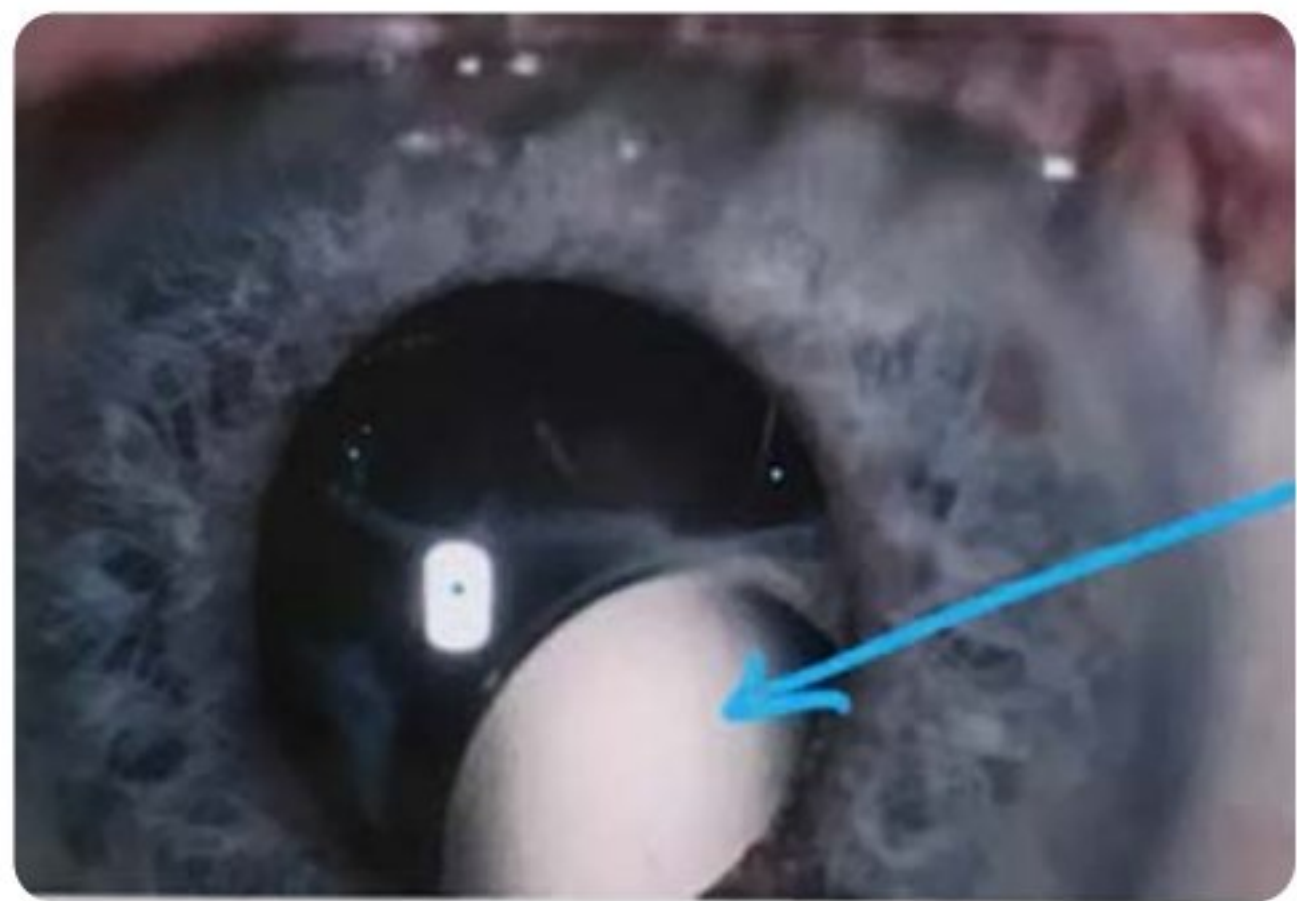
miotic pupil → Hits anterior capsule of lens → pigment ring

↳ made of melanin

9) Rosette Cataract → **Post. Cortical Cataract**



10) Posterior dislocation of LENS → vitreous

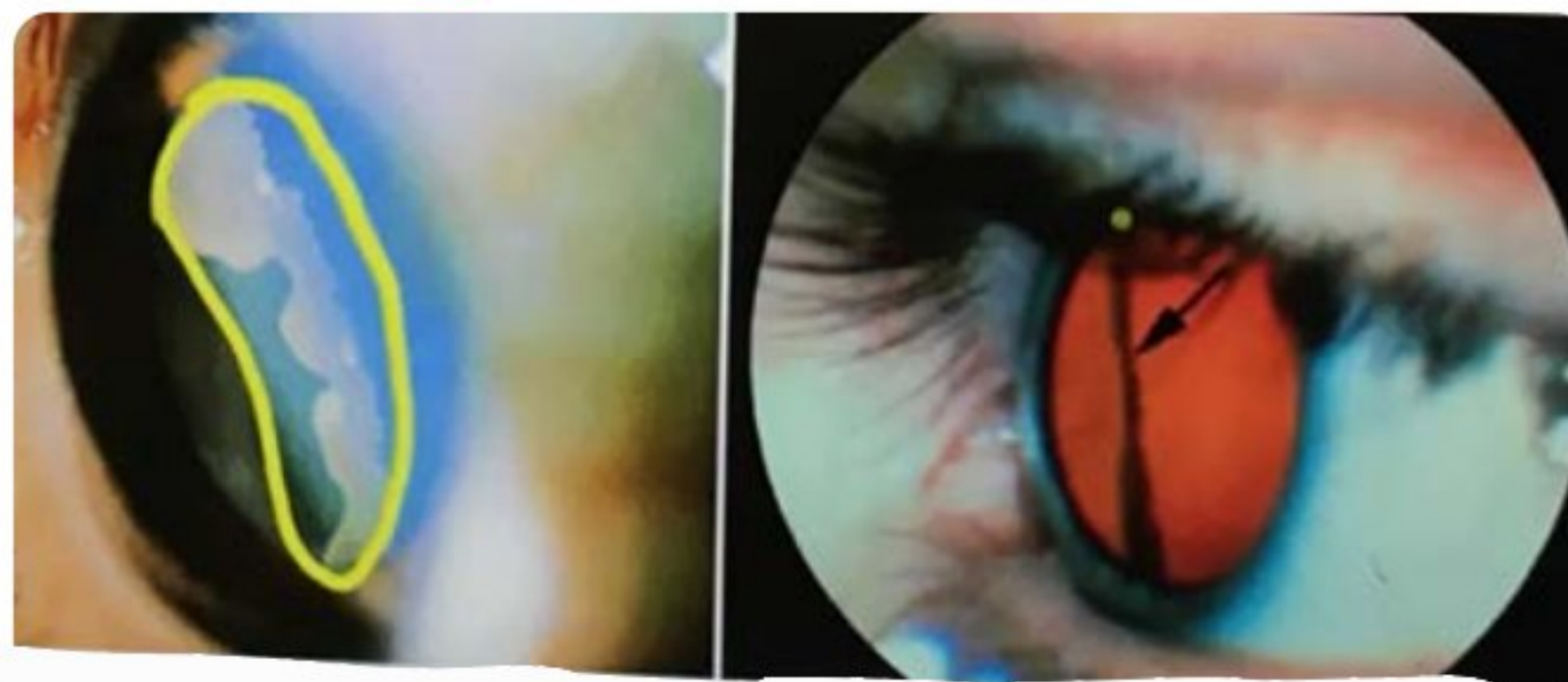
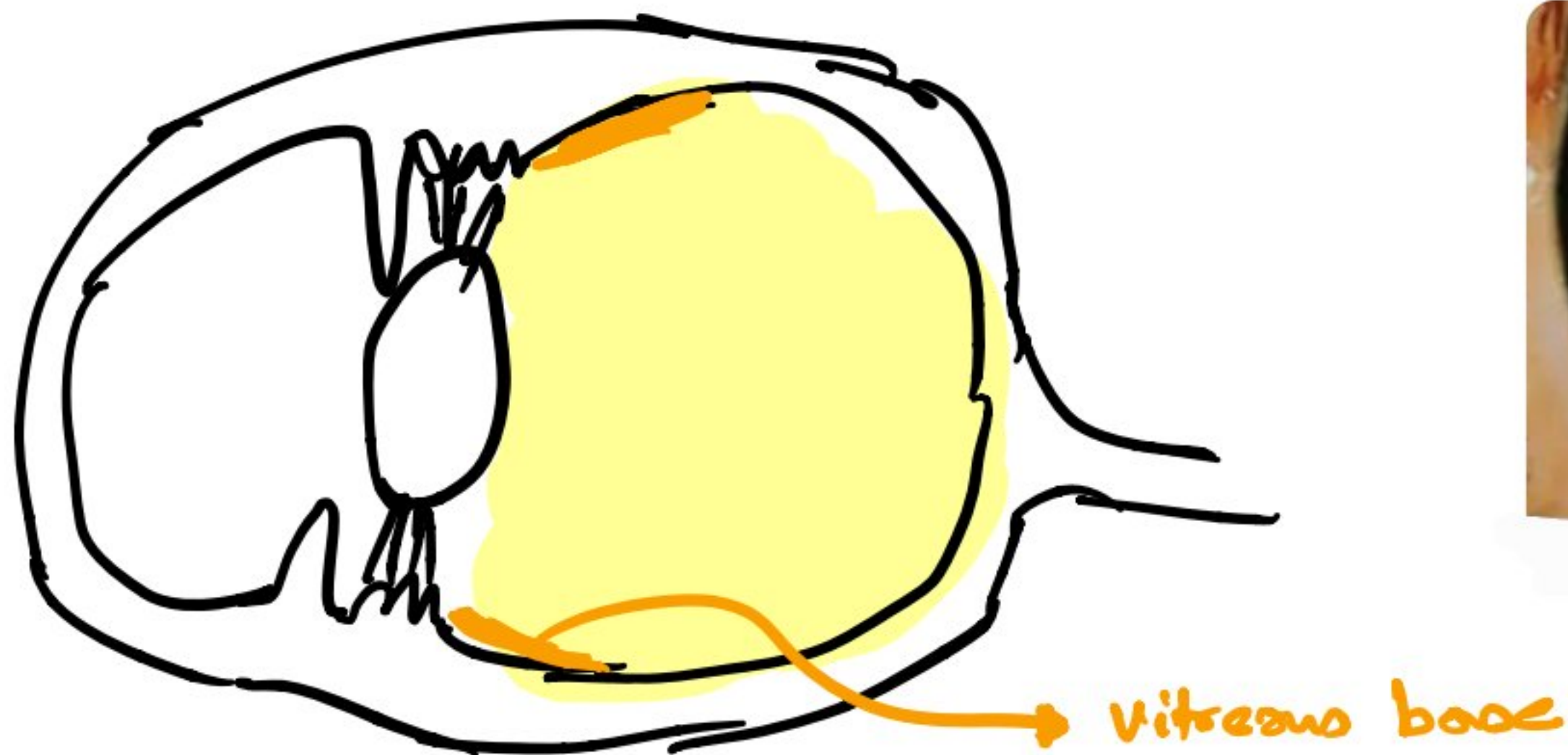


Q: Pt presented with sudden painless loss of vision following blunt trauma. Most likely cause?

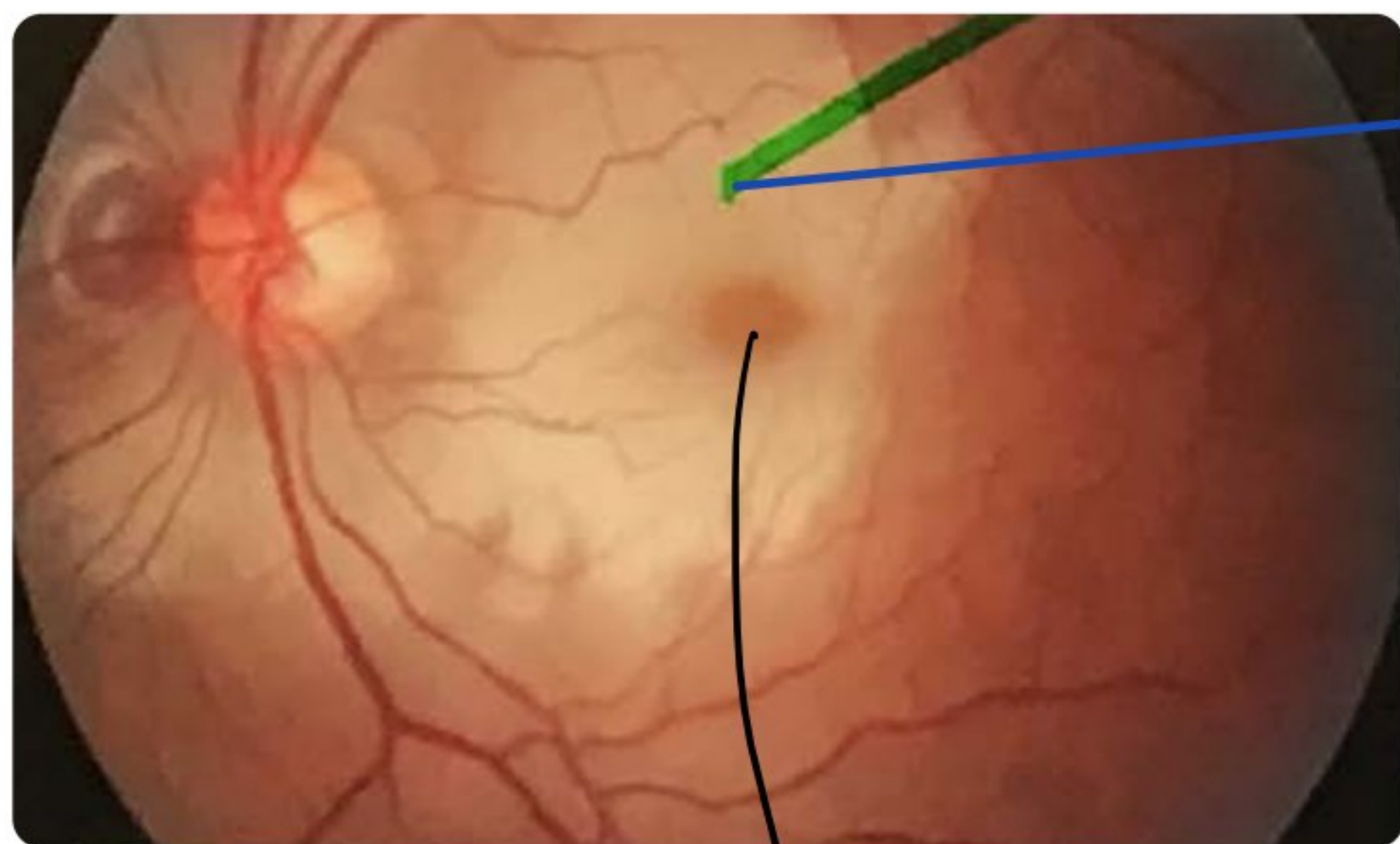
- a) Blunt trauma
- b) Iridodialysis
- c) Rosette Cataract
- d) ☒ Post. dislocation of lens

11) Vitreous H'mage → (MCC) in children

12) Vitreous base avulsion → PATHOGNOMONIC sign of blunt trauma.



13) Commotio Retinae : { Berlins edema + Cherry red spot



• milky white retina edema

Q: NEET PG

BERLIN'S EDEMA

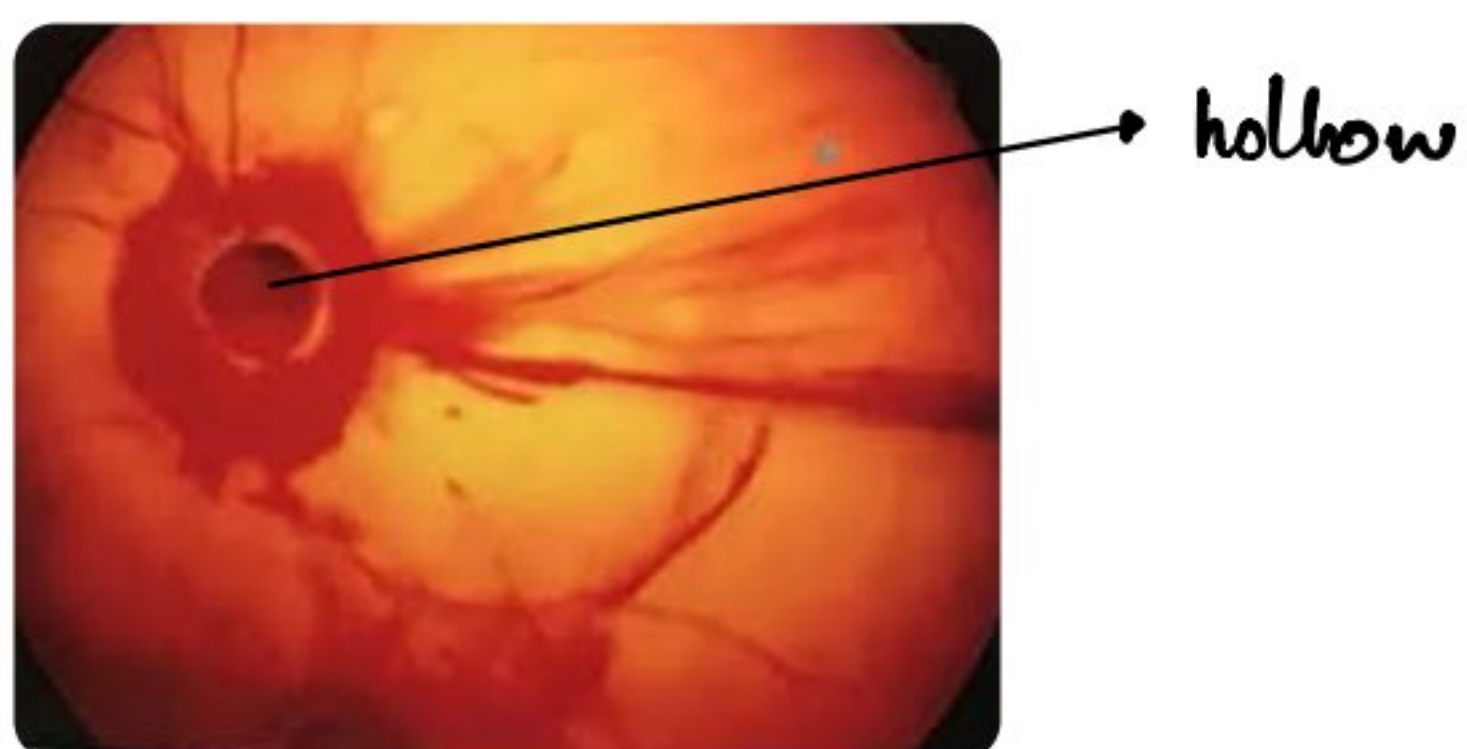
- ↑↑↑ IOP
- CRAO
- hypoxia of GCL
- Intracellular edema ? in ganglion cells
- ↓
- Retina opaque.

• Cherry Red spot

Q: Pt. presented with sudden painless loss of vision following blunt trauma. Most likely cause?

- a) ☒ Commotio Retinae → No vision
- b) Iridocyclitis
- c) Rosette Cataract
- d) Posterior Dislocation of lens → ↓sed Vision

14) Optic nerve Avulsion



TERSON'S Sx :

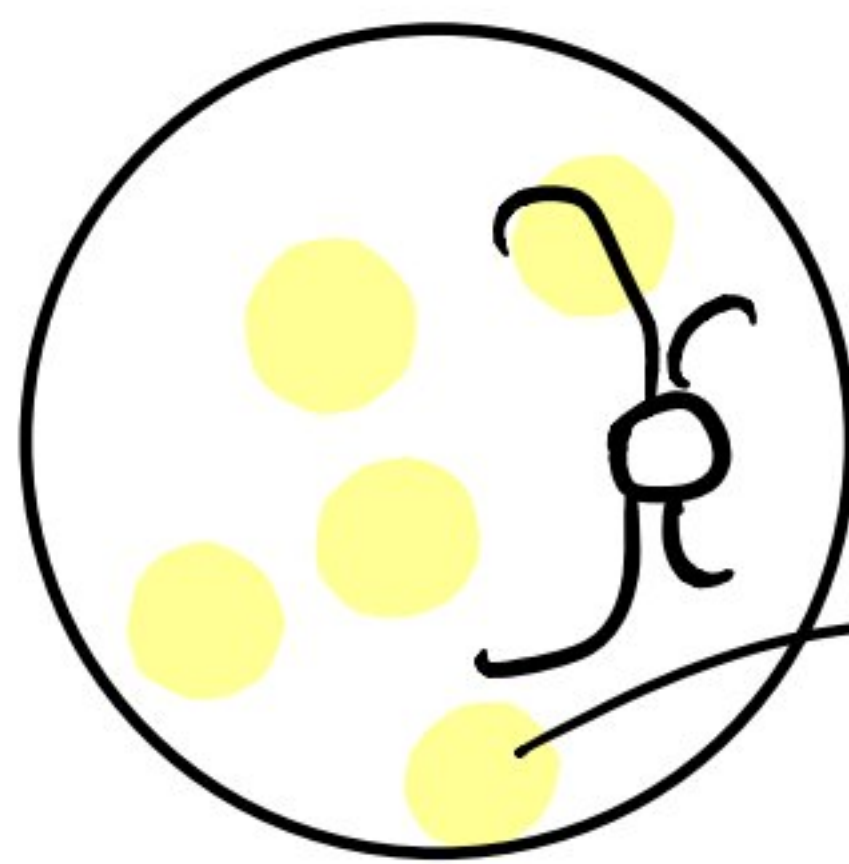
Intracranial H'mage $\begin{cases} \text{SAH} \\ \text{Intracerebral H'mage} \end{cases}$ { parenchymal $\rightarrow$ concussion }

⊕
Intraocular H'mage $\begin{cases} \text{VH} \\ \text{Pre retinal H'mage} \\ \text{Intra retinal H'mage} \end{cases}$

cause : ↑ sed CSF pressure.

PURTSCHER'S RETINOPATHY:

- Head injury
- Chest injury

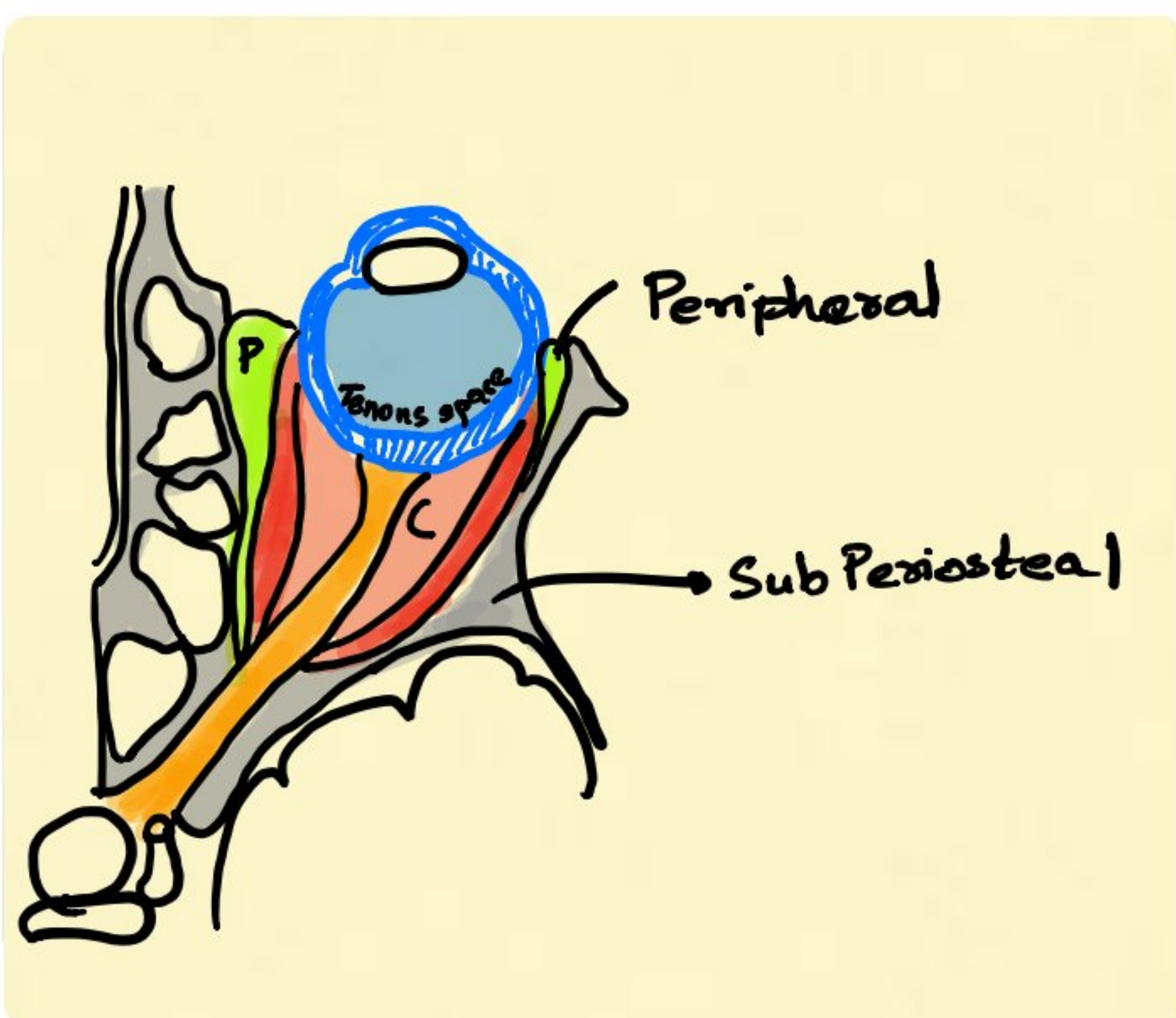
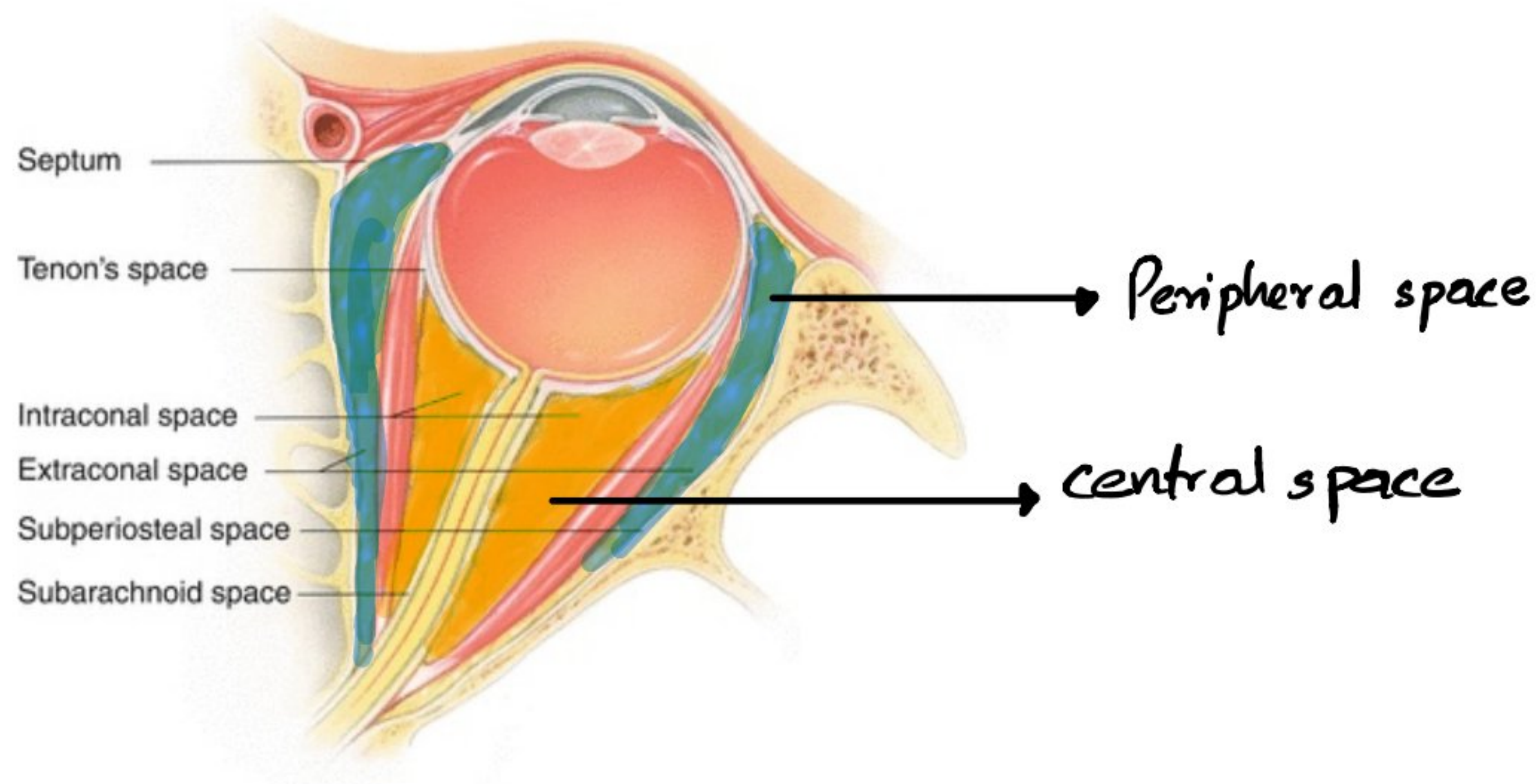


Fat Embolism

↓
RNFL ischemia

cotton wool spots $\rightarrow$ huge peripapillary in location

Surgical spaces in the eye :



PROPTOSIS ::

Instrument to measure proptosis is exo-ophthalmometer

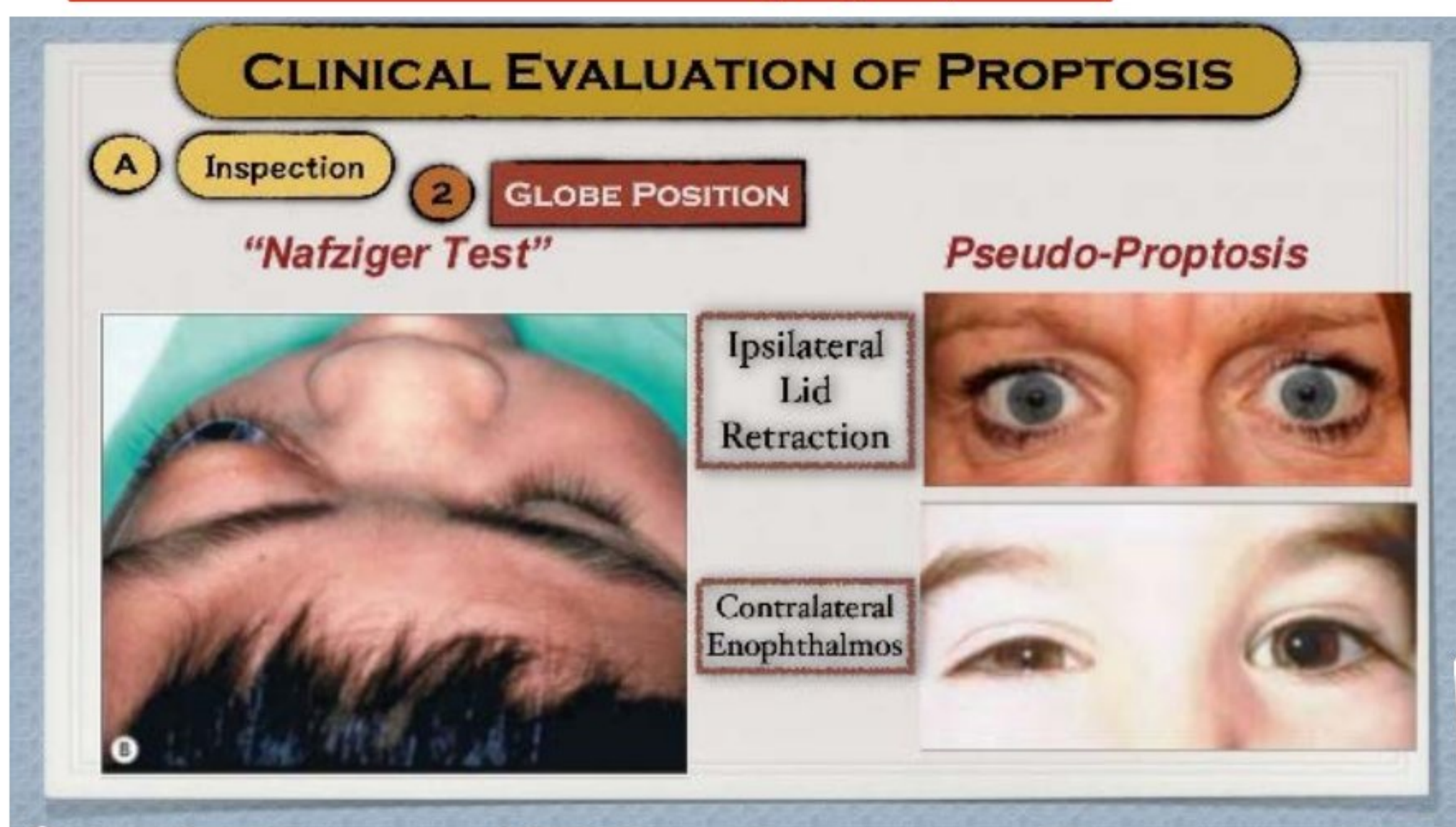
An absolute protrusion of $> 21\text{mm}$ or a difference of $> 2\text{mm}$ b/w two eye.

Hertel
Luedde
(for children)

Pseudo Proptosis

- 1) High axial myopia $\rightarrow AxL \geq 26\text{mm}$
- 2) C/L enophthalmos
- 3) Upper lid retraction
- 4) Buphthalmos & congenital Glaucoma

Clinical test to Identify proptosis



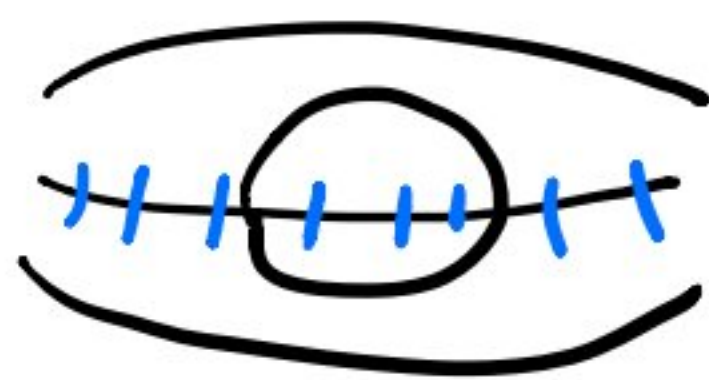
HERTHEL'S OPHTHALMOMETER



OBJECTIVE
 $> 21\text{mm}$
Triangle $> 2\text{mm}$

Complication of Proptosis ::

1) Exposure keratitis $\rightarrow$ TARSORAPHY



2) Diplopia

3) Optic N. compression $\rightarrow$ Proptosis $> 10\text{mm}$

- $\rightarrow$ i, ↓ visual acuity
- $\rightarrow$ ii, Ab(N) R, & colour vision
- $\rightarrow$ iii, RAPD

AXIAL PROPTOSIS

- Cavernous hemangioma
- Optic nerve glioma
- Thyroid eye disease
- Arteriovenous malformations, and
- Mass lesion within the muscle cone.



CAUSES OF ABAXIAL DISPLACEMENT OF GLOBE:

Downward displacement:

- fibrous dysplasia
- frontal mucocele
- subperiosteal hematoma
- lymphoma
- neuroblastoma
- neurofibroma
- schwannoma
- thyroid orbitopathy



Types

Axial

- Thyroid eye d/s
- Optic Nerve tumour
- Cavernous Hemangioma

Rx: Systemic Steroid

↓ If not resolve

Radiotherapy

↓ If not resolve

Orbital decompression ⇒

Medial wall > Inferior wall

> lateral wall > superior wall

Non axial

- Lacrimal gland tumour ⇒ down and nasal

MC → Benign Pleomorphic adenoma

M.C Malignant → Adenoid Cystic Ca.

MC non epithelial → Lymphoma

↓
Non Hodgkin Lymphoma (B cell type)

- Lacrimal sac ⇒ down & lateral.

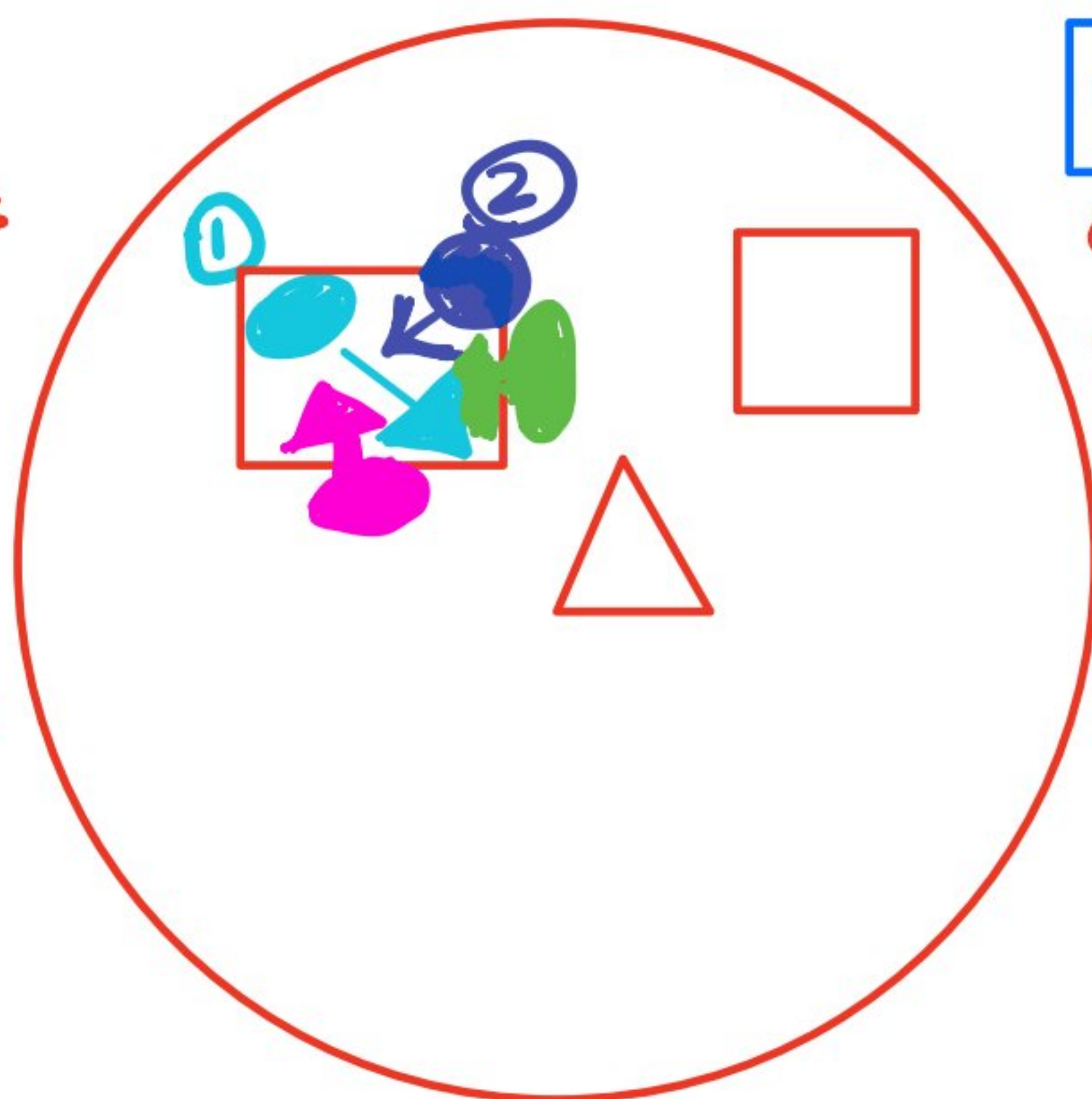
NON AXIAL / ECCENTRIC

Outward

Ethmoid sinusitis

UPWARD

Ca. Maxillary sinus
↑ orbital spread



DOWN & MEDIALY

- ① LG tumor
- ② Dermoid Cyst

DOWN & OUT

- ① Frontal Sinus mucocele
- ② Meningoencephalocele

Proptosis

{ Based on Duration }

Ac

- Orbit H'mage
- Orbital cellulitis

Intermittent

- Orbit varix
- Orbit lymphangioma

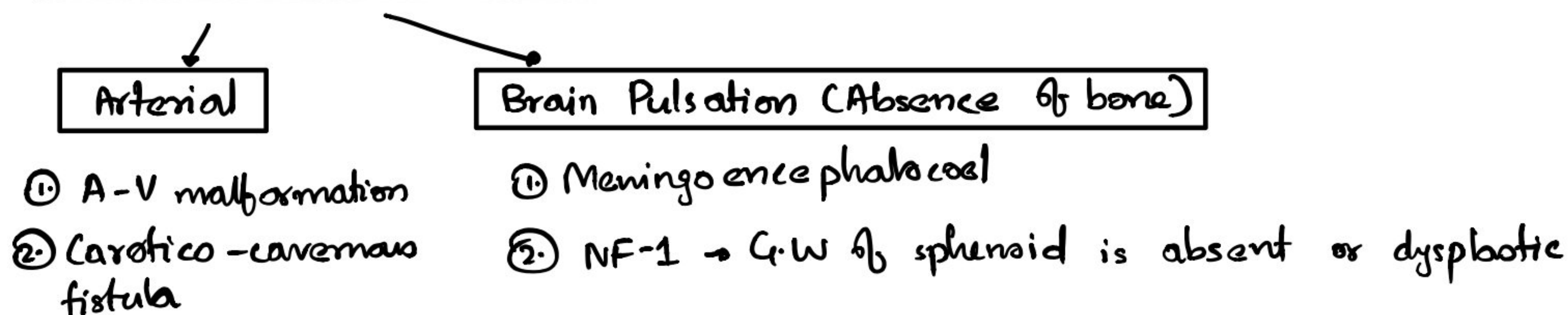
Pulsating

- Carotico cavernous fistula
- Meningoencephalocele

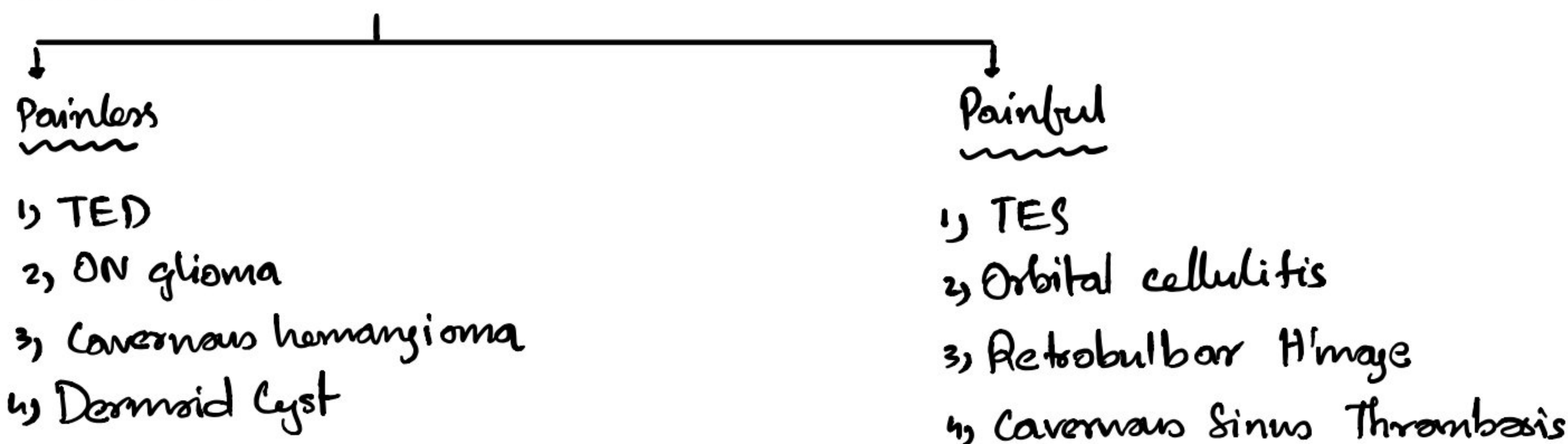
Intermittent Proptosis

- i) Orbital Varices: Valsalva manoeuvre → venous pressure ↑
venous varices ⊕
- ii) Orbital Lymphangioma: URTI ⇒ Lymphoid tissue hyperplasia

Pulsating Proptosis (Transmitting)



Proptosis (Based on pain)



Unilateral proptosis, pain, fever, decreased ocular motility, erythema and edema of the eyelids

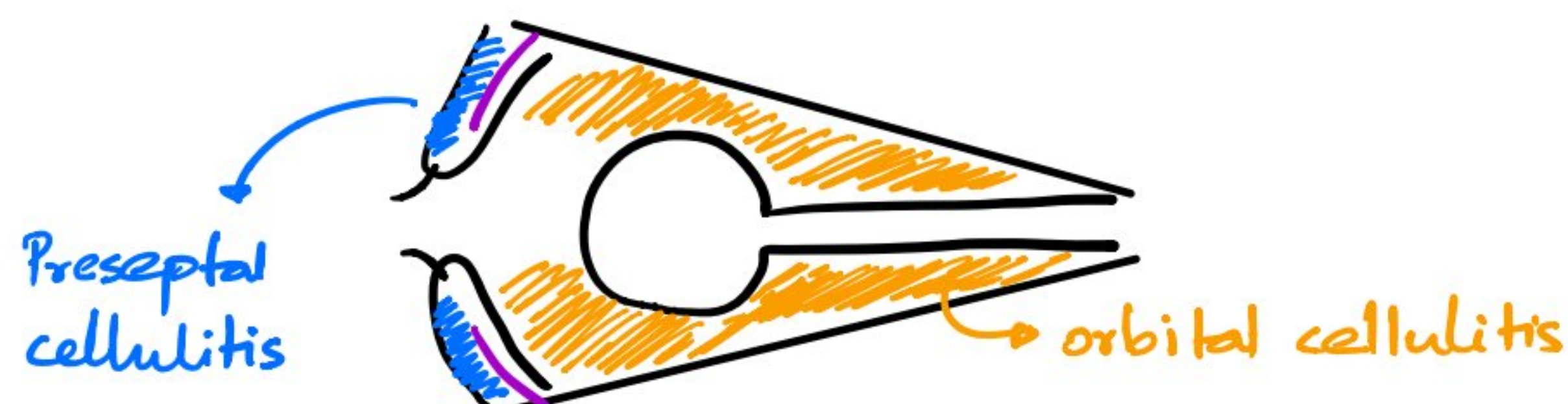


→ Subperiosteal Abscess

→ ORBITAL CELLULITIS
d/t ethmoidal

Orbital Cellulitis:

- Painful proptosis ; fever
- Usually bacterial
- Inflammation extends posterior to orbital septum



ⓂⓈ source → Ethmoid sinusitis
 Rx: 1-2 antibiotics

O.C	Preseptal Cellulitis
V. Acuity ↓se	V. A (N)
Colour vision ↓se	C. V (N)
RAPD	(N)
Limitation of eye movement → diplopia	Eye movement → (N)

Complication:

- 1) Compressive optic neuropathy → Blindness d/t Optic Atrophy
- 2) Cavernous Sinus Thrombosis
- 3) B/L Orbital cellulitis ← ⓂⓈ
- 4) Cranial Neuropathy

↳ @ SOF : III, IV, VI, corneal sensation ⊖

@ CST : VI, III, IV, V₁, V₂

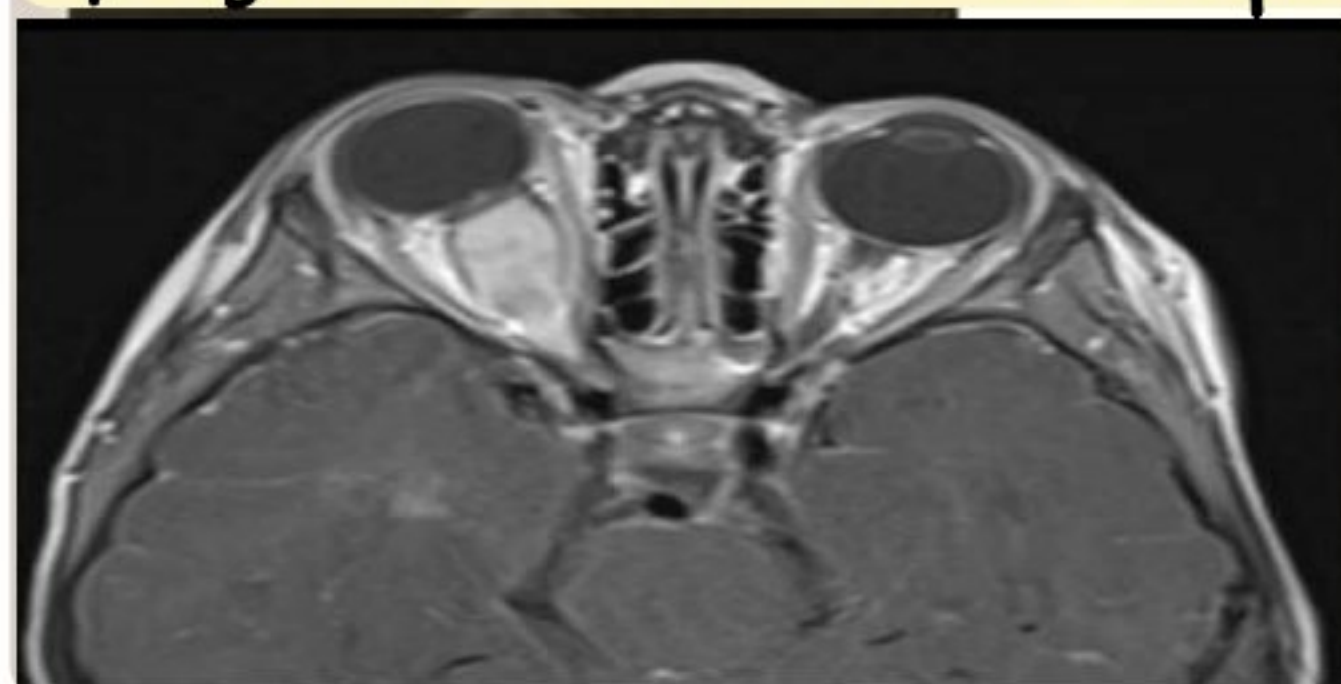
CHANDLER CLASSIFICATION OF ORBITAL CELLULITIS

- 1) Preseptal cellulitis
- 2) Orbit cellulitis
- 3) Subperiosteal abscess
- 4) Orbital abscess
- 5) Cavernous Sinus Thrombosis

Ⓢ A patient with Ⓡ orbital cellulitis is not responding to antibiotics and is seen to be worsening after initial presentation. The first sign to raise the suspicion of involvement of the cavernous sinus is:

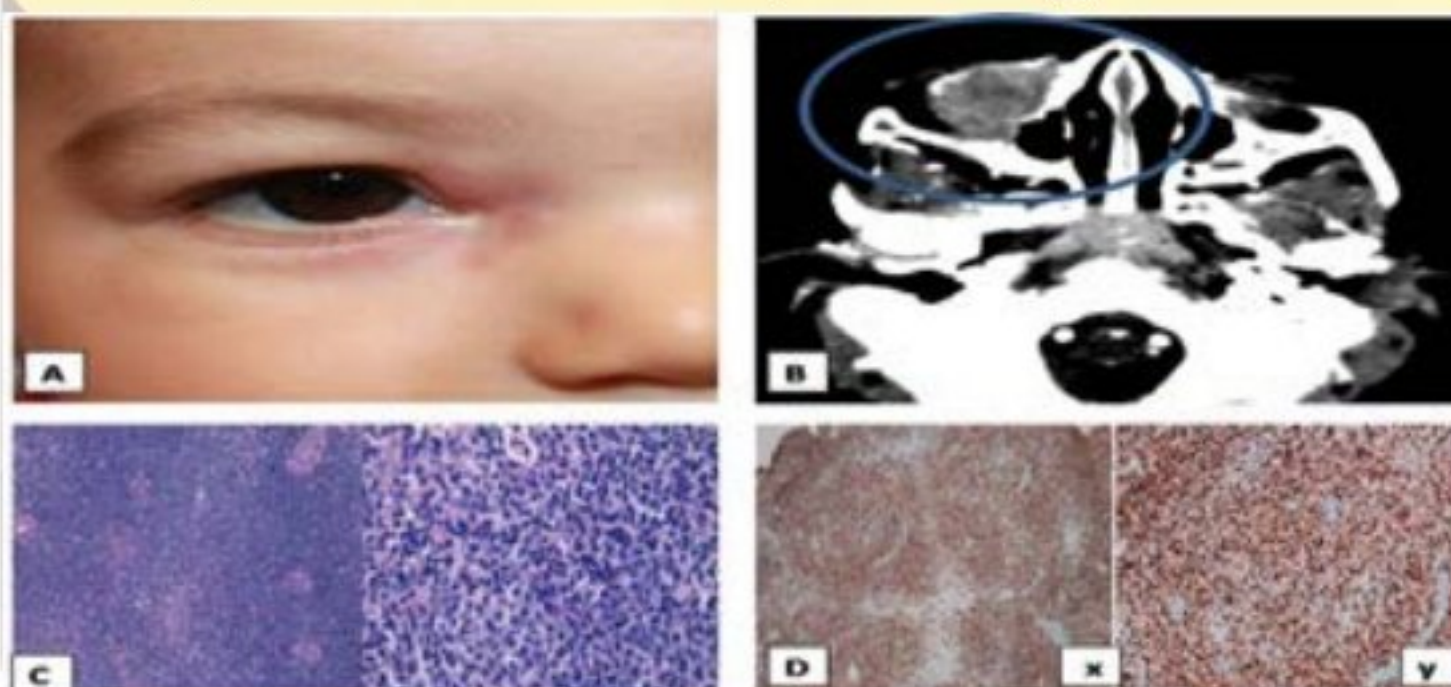
- a) Painful right Ophthalmoplegia
- b) Paralysis of Ⓡ superior oblique
- c) Paralysis of Ⓡ Lateral rectus → C/L CST
- d) Anaesthesia of Ⓡ cornea

child < 6yrs with gradual, painless, progressive unilateral axial proptosis + visual loss



R. Optic Nerve glioma

child with rapidly progressive U/L proptosis displacement of the globe inferiorly and edema



Rhabdomyosarcoma

RHABDOMYOSARCOMA

- (MC) site = superior orbit
- Arise from pluripotential mesenchymal precursor
- Age = 7yr
- Dx = Incision Biopsy
- Rx = RT + CT
- Px = Mortality 60%
- (MC) type = Embryonic
- worst type = Alveolar 2nd (MC)

- (MC) 1° malignant intraorbital in child = Rhabdomyosarcoma
- (MC) Intraorbital tumour in child = Orbital mets from Neuroblastoma



- B/L proptosis
- B/L periorbital H'mage = Raccoon eyes

- (MC) Intraorbital tumour in children → **DERMOID CYST**

InxOC: CT orbit
= dumbbell tumour

- (MC) intraorbital tumour in adults → cavernous hemangioma
- (MC) malignant " " " → **Orbital mets** from
 - ♀ = Ca Breast
 - ♂ = Ca. Prostate
 - smoker = Ca. Lung

↑ Enophthalmos

SEROUS Type of Ca. Breast

- (MC) 1° malignant intraorbital tumour in adults → B cell NHL (Diffused large cell)

- (MC) location = superior orbit

Salmon flesh app = subconjunctival area

Rx = Chemotherapy

- (MCC) of B/L / U/L Proptosis → Thyroid eye d/s
- Exophthalmos → Proptosis in thyroid eye d/s

Thyroid eye d/s:

Pathogenesis

Orbital Fat inflammation



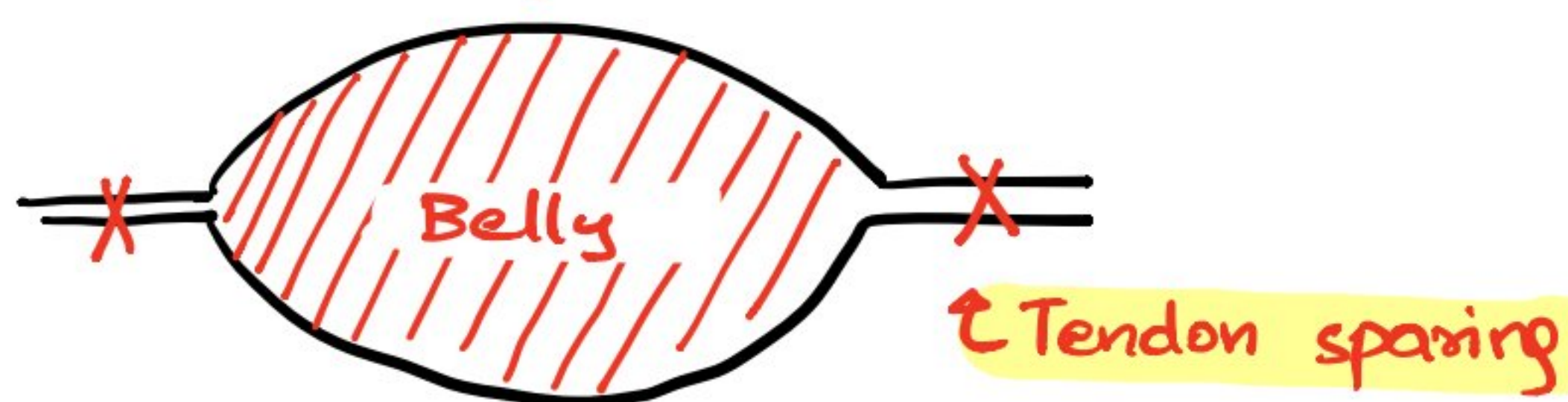
Fibroblast = infiltration → Fat
→ Edema



Glycosaminoglycan deposition (GAG)

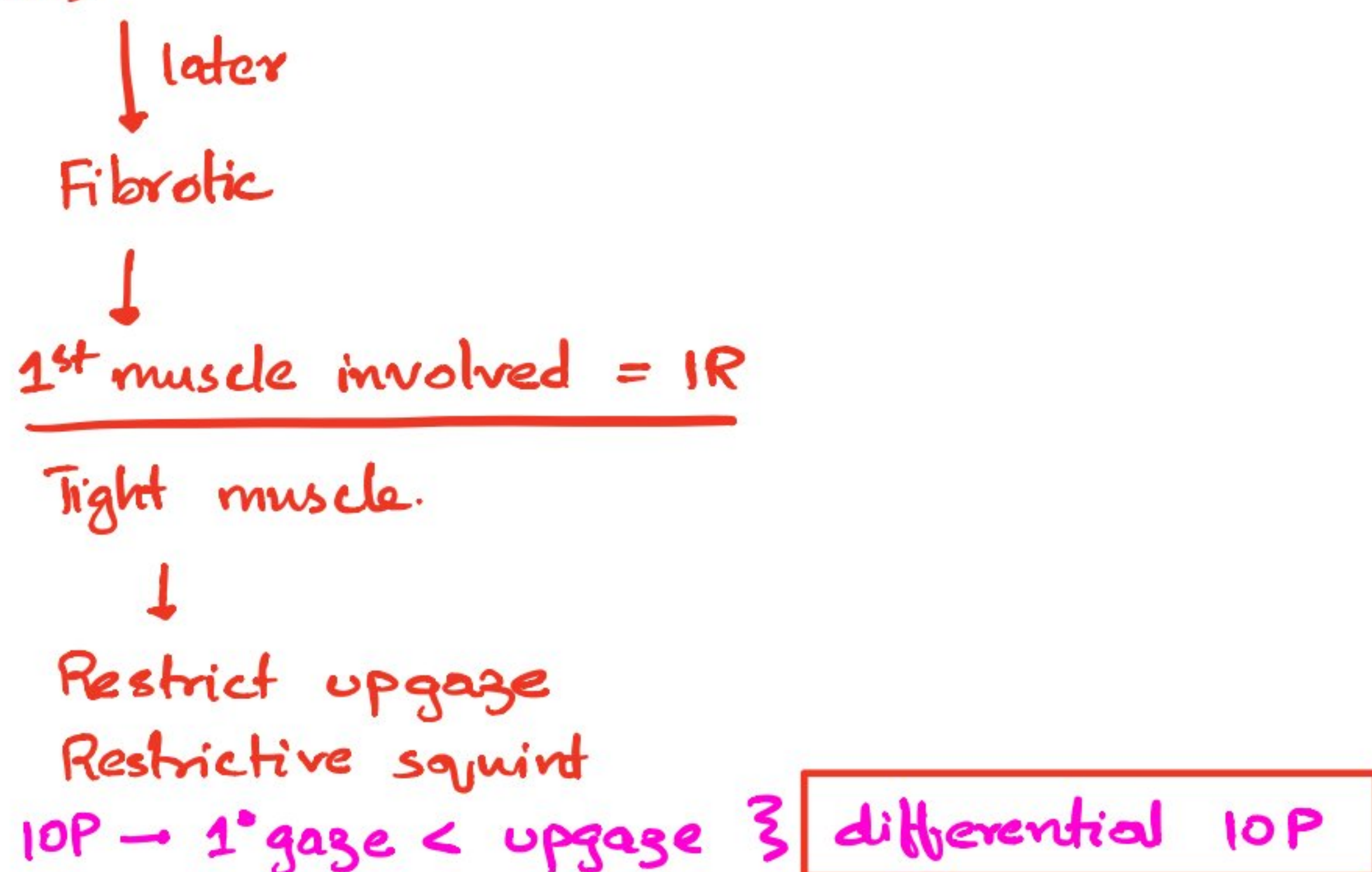
Hydrophilic

EDM swelling → Fibrotic



- (MC) muscle involved
I > M > S > L rectus
- Least muscle involved → IO
- (MC) action involved → Elevation

EDM belly - enlargement (edema)



- 1) F > M
- 2) Painless
- 3) Smoking ↑ risk by 5 times
- 4) ↑ orbital inflammation
- 5) ↑ GAG deposition → absorb water → swell

Thyroid eye sign:

- 1) (MC) / 1st sign → Lid retraction → **Dalrymple sign**
- Proptosis
- ↑ Muller's action → 1-2 mm eyelid elevation
d/t ↑ sympathetic activity.

- 2) Starting look ← **Kocher's sign**
- 3) Infrequent blinking ← **Stellwag's sign**
- 4) Lid lag sign ⊕ ← **Von Graefe's sign**
- 5) Enophthalmos = puffiness of eyelid

Class 0 → No physical sign & symptoms

- 1 → Only signs, no symptoms
i) Dalrymple ⊕
ii) Von Graefe's ⊕
iii) Stellwag ⊕

- 2 → Soft tissue involved → signs
→ symptoms

- 3 → Proptosis

- 4 → EDM involved

- 5 → Cornea involved

- 6 → Sight loss & optic N. involved

Rx: Euthyroid

↓
Systemic steroids

↓
Immunosuppression

↓
R.T

6) Moebius sign = Weakness of convergence. {last & most severe sign}

DALRYMPLE SIGN



MOBIUS SIGN



KOCHER'S SIGN



VON GRAFFE'S SIGN



7) Joffrey's Sign → Lack of wrinkling of forehead on superior gaze.

8) Gibbard sign → difficulty in eversion of the upper lid.

Optic Nerve tumours:

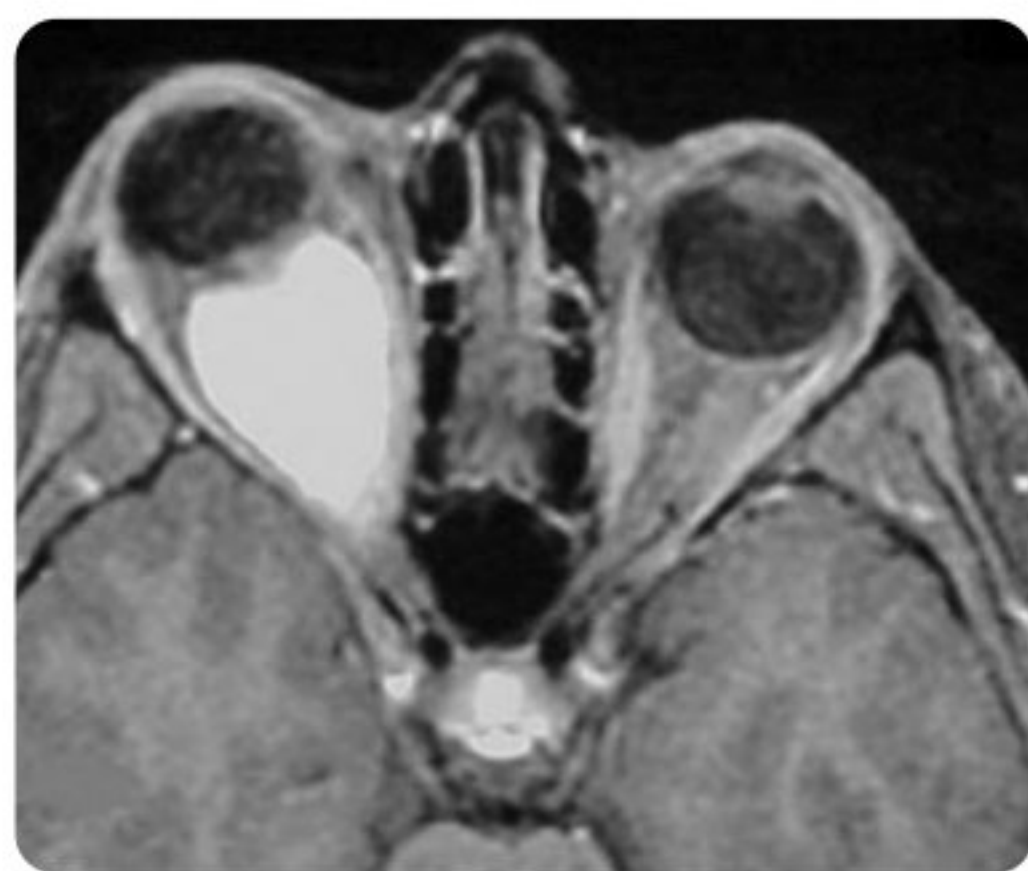
Glioma

Painless, Axial
Proptosis

F > M, 5-10yr

Fusiform app.

Pilocytic astrocytoma



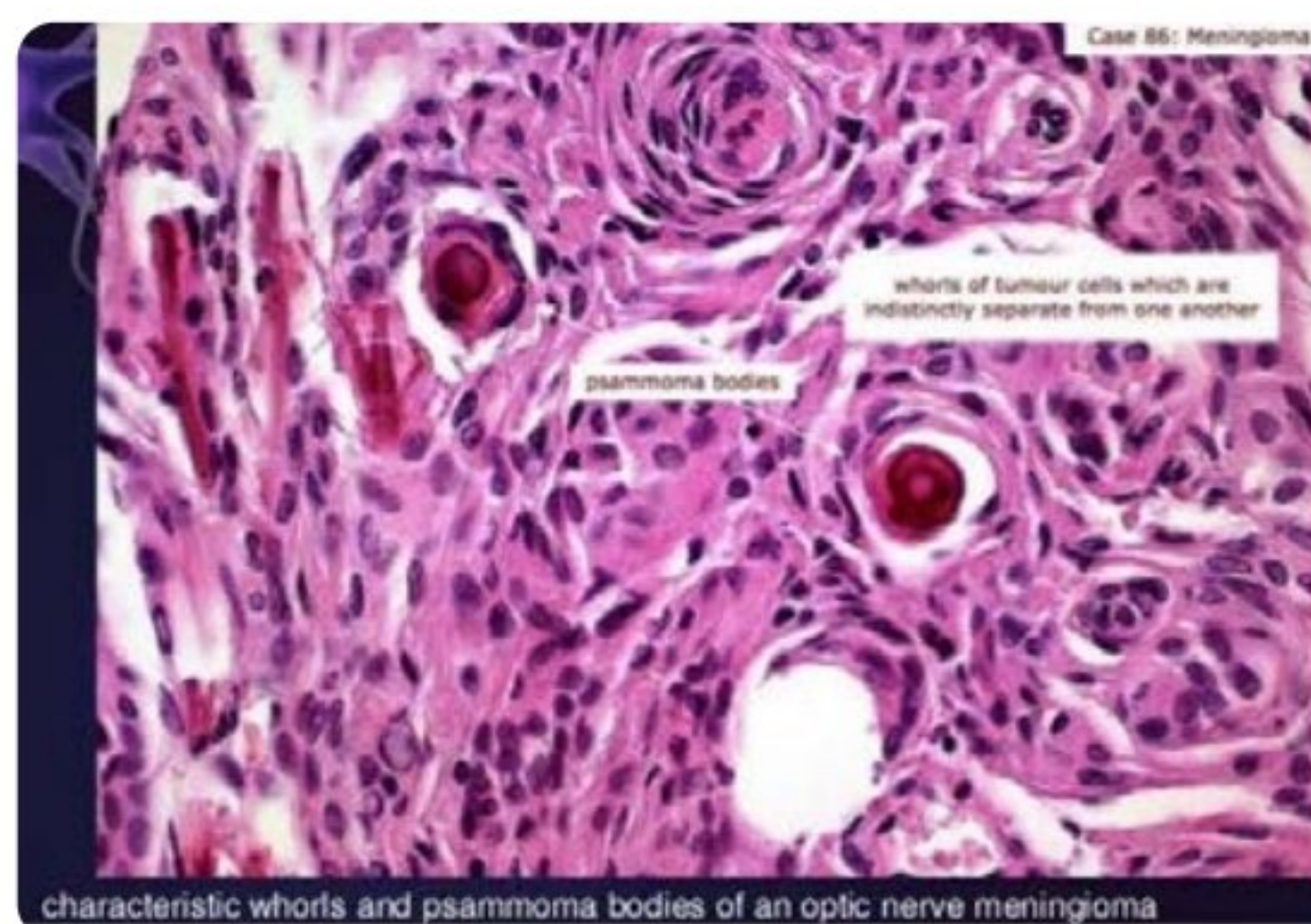
Optic Nerve glioma

Meningioma

F > M, middle aged

Tubular Calcification

Psammoma Bodies



characteristic whorls and psammoma bodies of an optic nerve meningioma

a/w NF-1

↳ Plexiform Neurofibroma

⊕ [S-shaped eyelid]

↳ Lisch nodules (Hamartoma)

↳ Sphenoid dysplasia

Other features of Optic Glioma:

• (MC) type → Juvenile pilocytic Astrocytoma } Rosenthal fibres

⊕ All of the following are treatment modalities for thyroid ophthalmopathy except

- a) Radiation
- b) Steroids
- ✓ c) **B Blocker**
- d) Orbital decompression

⊕ NF-2 (MC) ocular feature → post. Subcapsular cataract

⊕ (MC) U/L proptosis in children → orbital cellulitis | B/L proptosis → orbital mets from neuroblastoma

⊕ (MCC) → Ethmoid sinusitis

⊕ (MC) Bacteria → Staph Aureus/ Haemophilus

⊕ (MC) Fungus → Mucor

⊕ (MC) Parasite → Cysticercosis

Rx $\Rightarrow$ Systemic AB

(MCC) B/L proptosis in children
Neuroblastoma > ALL

PROPTOSIS

Pulsatile

• Carotico-cavernous fistula

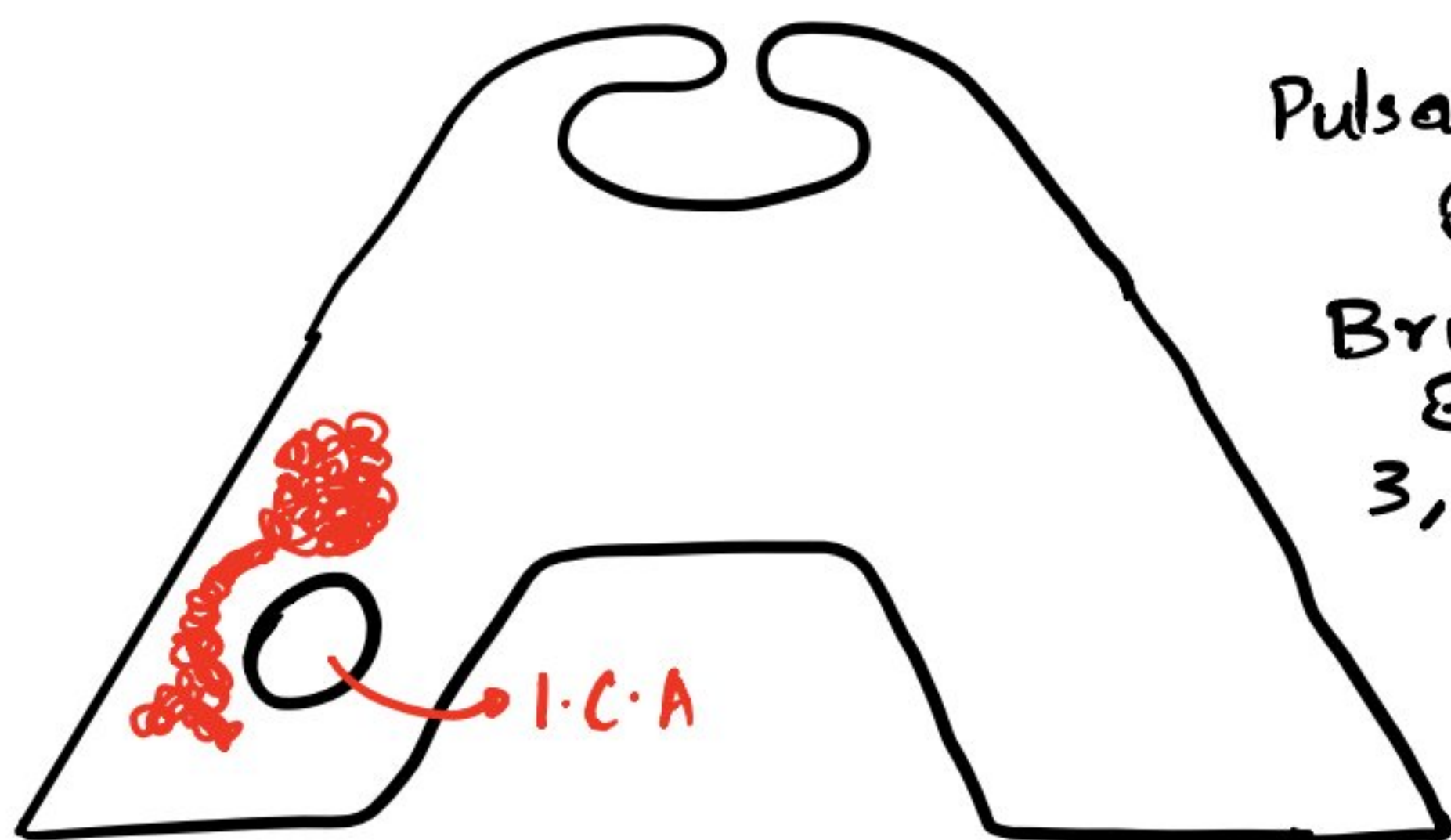
• roof of orbit
• sphenoid dysplasia
• meningocoele

} Transmitted pulsat

Q: A 26 yr old man with a history of Blunt trauma to the head 2 months earlier presented with rapidly progressive swelling pain, and diminishing vision of the $\textcircled{L}$ eye. The physical examination of the $\textcircled{L}$ eye revealed chemosis and pulsating proptosis. A bruit could be heard with a stethoscope placed on the $\textcircled{L}$ temple. The most probable Dx is ?

- a) Carotid - cavernous fistula
- b) Retro-orbital hematoma
- c) Superior orbital fissure syndrome
- d) Orbital apex syndrome.

Carotico - Cavernous fistula : ICA $\rightarrow$ cavernous sinus communication
d/t head trauma or atherosclerosis (old)
(young)



Pulsatile properties

⊕

Bruit

⊕

3, 4, V₁, V₂, 6

+/- Horner

Proptosis $\rightarrow$ With thrill

⊕

murmur

• $\uparrow$ episcleral venous pressure

↳ $\uparrow$ IOP $\rightarrow$ 2° glaucoma $\rightarrow$ AQ drain x

↳ Cork screw vessels

• c-c fistula a/w • Thyroid d/s

• Sturge-Weber syndrome $\Rightarrow$ 50% have glaucoma

Intermittent

• Orbital varices $\rightarrow$ veins of orbit. thin

• young males

• $\uparrow$ on coughing

bending

Valsalva Manouever



Rx : Excise vein

or add sclerosants

• Bruit is -ve



Chemosis - cork screw vessels

Q: A 20yr old man with a h/o of repeated pain over the medial canthus and C/C use of nasal decongestants presents to the outpatient department with high-grade fever and chills, headache and B/L proptosis. On eliciting further history, it is brought to light that the proptosis began in the (R) eye, and a few days later, the (L) eye was also affected. Which of the following is the most probable diagnosis?

a) Orbital Cellulitis

☒ b) Cavernous Sinus Thrombosis

c) Panophthalmitis

d) None of the above

- In CST, proptosis is initially U/L but soon becomes B/L due to the spread of infection to the contralateral sinus through intercavernous communication.
- High grade fever with chills, rigor, and headache.

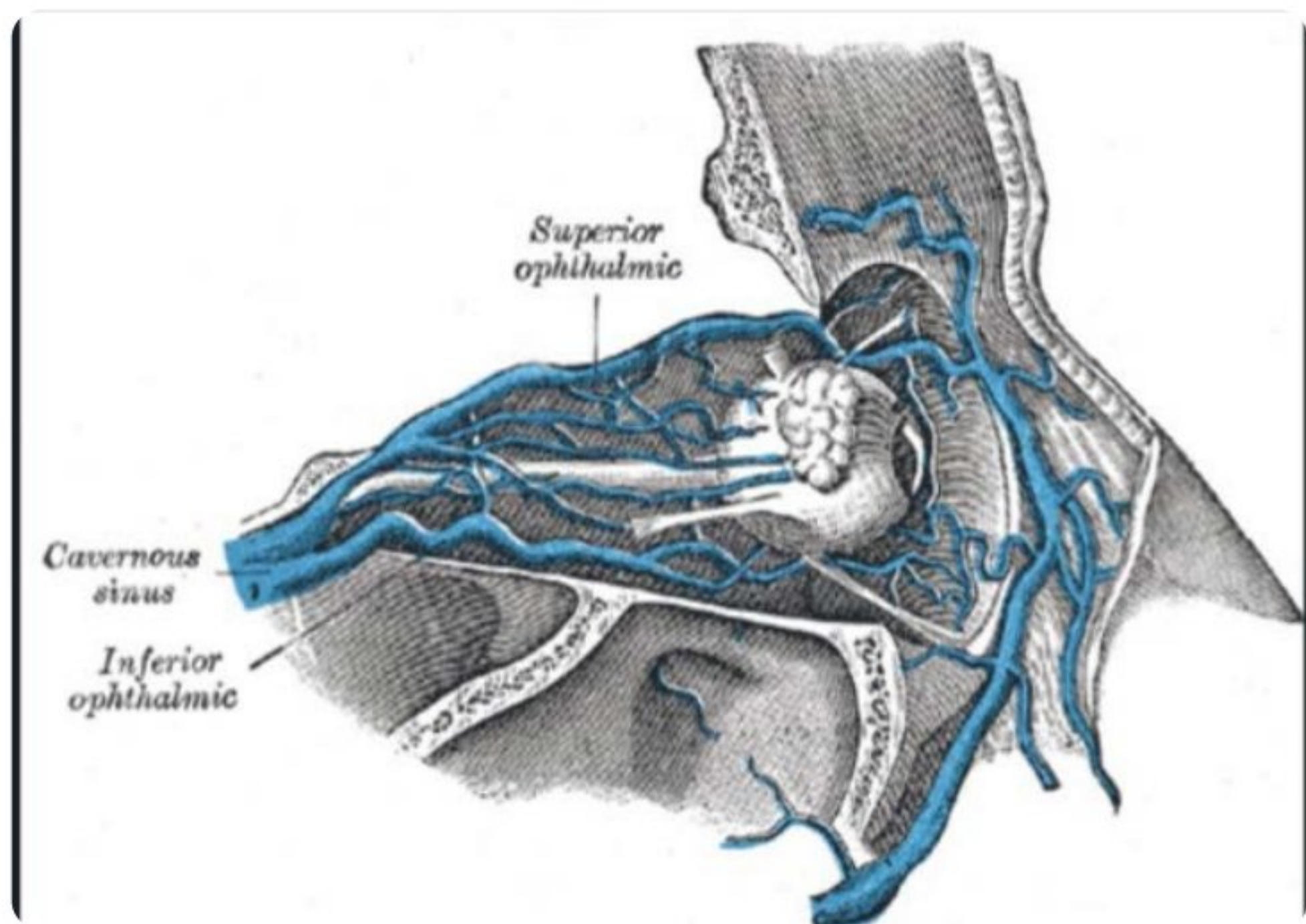
Clinical feature	Cavernous sinus Thrombosis	Orbital cellulitis	Panophthalmitis → Inflammation of whole eye along with Tenon's capsule
Laterally	Initially U/L but soon becomes B/L	U/L	U/L
Vision	Not affected in early stage	affected	Complete loss of vision
Cornea and Ant. chamber	Clear	Clear	Hazy
Edema in the mastoid region	Present (Pathognomic of cavernous sinus thrombosis)	Absent	Absent
General symptoms with fever and prostration	Marked	Mild	Mild

Q: Infection from the dangerous area of the face spreads to the cavernous sinus via the _____?

- a) Maxillary veins
- b) Retromandibular veins
- c) Super temporal vein

~~d) Ophthalmic veins~~

• Infection from the dangerous area of the face spreads to the cavernous sinus via the ophthalmic veins



Q: The surgery of choice for panophthalmitis is

~~a) Evisceration~~

- b) Enucleation
- c) Exenteration
- d) Debulking

• Evisceration :: removal of the contents of the eyeball, leaving behind the sclera

- panophthalmitis
- evisceration is done

• Enucleation :: removal of the eyeball with the max^m part of the optic nerve

- extraocular muscles are left behind
- Length of the optic nerve removed is 10 to 15mm.
- Absolute indications for enucleation are retinoblastoma and malignant melanoma.

• Exenteration :: is a mutilating surgery in which all the contents of the orbits along with the periorbital are removed.

- It is indicated for malignant tumours arising from the orbital structures or spreading from the eyeball.

Debulking :: reduction of as much of the bulk (volume) of a tumor as possible.

- Nowadays, debulking of the orbit is preferred over exenteration

Q. The syndrome associated with hypoplasia of the zygoma and mandible is
a) Crouzon Syndrome

b) Treacher Collins syndrome

c) Apert Syndrome

d) Median facial cleft syndrome

sol. Syndrome associated with hypoplasia of the zygoma and mandible is
TREACHER - COLLINS SYNDROME.

Treacher - Collins syndrome / Mandibulofacial dysostosis refers to a condition resulting from the hypoplasia of the zygoma and mandible

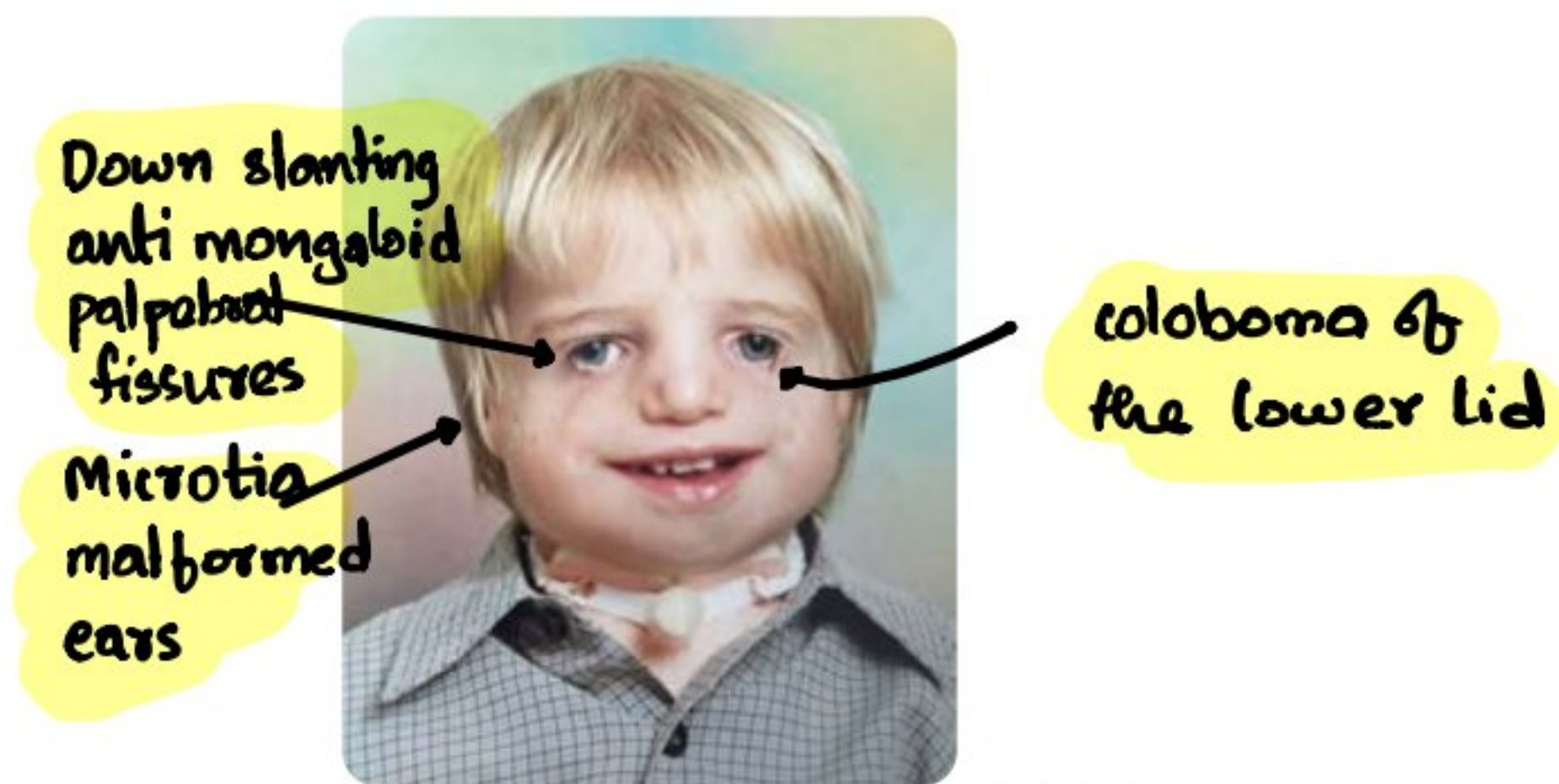
Ocular features:-

- Indistinct: inferior orbital margin
- Coloboma of the lower eyelid
- Anti-mongoloid slant

Systemic features:-

- Macrostomia & High arched palate
- External ear deformity
- Bird like face

Features of Treacher - Collins syndrome



★ WATCH THE FILM WONDER

Craniofacial dysostosis / Crouzon syndrome

↳ Premature closure of coronal & sagittal sutures (brachycephaly - clover-leaf skull) associated & maxillary hypoplasia.

Ocular associations include the following:-

- Blue sclera
- Megalocornea
- Aniridia
- Glaucoma
- Cataract
- Ectopia Lentis
- Optic Nerve hypoplasia

Apert Syndrome :: severe form of craniosynostosis

- premature closure of multiple cranial sutures (acrocephaly), along with syndactyly of fingers and toes

BLOW OUT #

Q. All the following statements are true regarding a "blow-out" # except that

- a) Anaesthesia of the cheek is very common
- b) Emphysema of eyelids is more common with medial wall #
- c) Enophthalmos develops about 10 days after the injury

d) Severe ocular damage is s/i most cases

Q: Which of the following views in an X-ray scan helps in the visualization of the optic foramina?

- a) Caldwell views
- b) Water's view
- c) Lateral view
- d) Rhese view

X-RAY views of the orbit

1) Caldwell's view : Occipito frontal.

Structure seen :

- a) Superior orbital fissure
- b) Greater and lesser wing of sphenoid
- c) Ethmoid and frontal sinus
- d) Floor of the orbit

2) Water's view

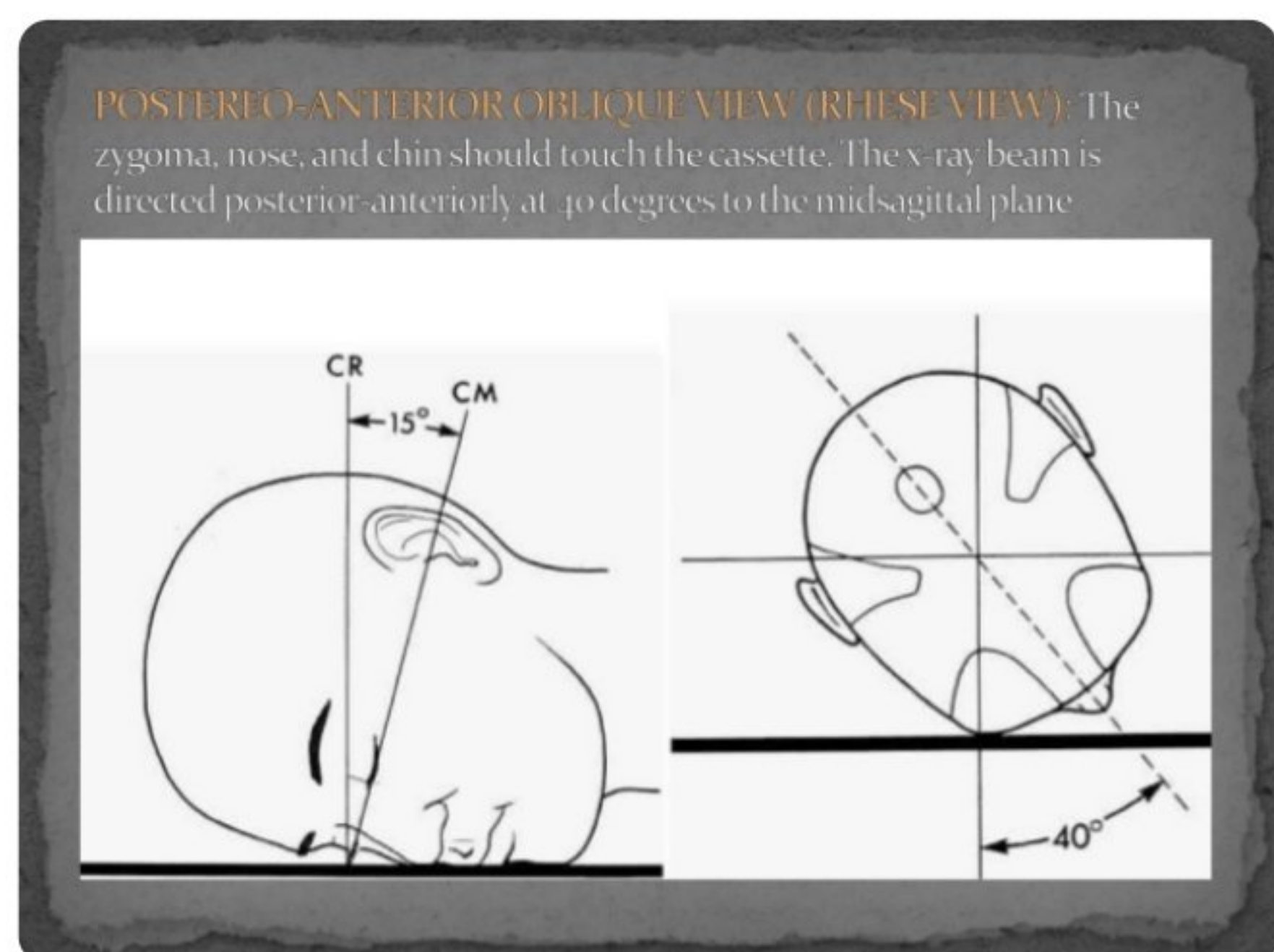
- a) Occipitomental : floor of the orbit is seen (important in blow out #)
- b) Occipitomental : maxillary sinus and floor of the orbit

3) Towne's view

- a) Dorsum sellae of the sphenoid bone
- b) Inferior orbital fissure

4) Rhese view :

For optic foramen



Q: Heterochromia iridis may be caused by which of the following intraocular foreign bodies?

- a) Copper
- b) Iron
- c) Aluminium
- d) Glass

- Siderosis bulbi: refers to the ocular degenerative changes produced by an. iron foreign body
 - greenish → reddish brown
 - Heterochromia Iridis

Chalcosis: produced by the alloy of copper in the eye.

CF:

- Kayser - Fleischer ring: it is a golden brown ring that occurs d/t the deposition of copper in the Descemet's membrane of the cornea
- Sunflower cataract: It is produced by the deposition of copper under the anterior capsule of the lens. It is brilliant golden green in colour and arranged like the petals of a sunflower



BLINDNESS: vision $< 3/60$ in better eye with available correction / central visual field $< 10^\circ$.

1999 → WHO + N40 → Geneva

VISION 2020 { Right to sight } ⇒ eradicate MCC of blindness

- i, Cataract
- ii, Refractive error
- iii, Childhood blindness
- iv, Trachoma
- v, Onchocerciasis (River Blindness)

Target: To reduce population of blindness to $< 0.3\%$.

Indian modification → 2004 at Goa.

Vision - 2020 India

- i, Cataract (MC)
- ii, Refractive Error
- iii, Childhood blindness
- iv, Corneal Blindness
- v, Glaucoma
- vi, Diabetic Retinopathy

Neet pg

- In India currently criteria $< 3/60$ (WHO also)
- Total blind pop. in India, 2019 = 8.45%.
- MCC of reversible blindness → Childhood → w/ A deficiency
- Irreversible → Glaucoma.

• MC cause of blindness → **Cataract**.

• MCC of **infectious cause** of preventable blindness → **Trachoma**.

Rashtriya Bal Swasthya Karyakram (RBSK)

Aim: Screening
early detection
Free management } of 30 selected health conditions

Age: Birth to 16 years

Eye condition covered:

- i) Vit. A deficiency
- ii) Congenital cataract
- iii) Vision impairment
- iv) Retinopathy of prematurity

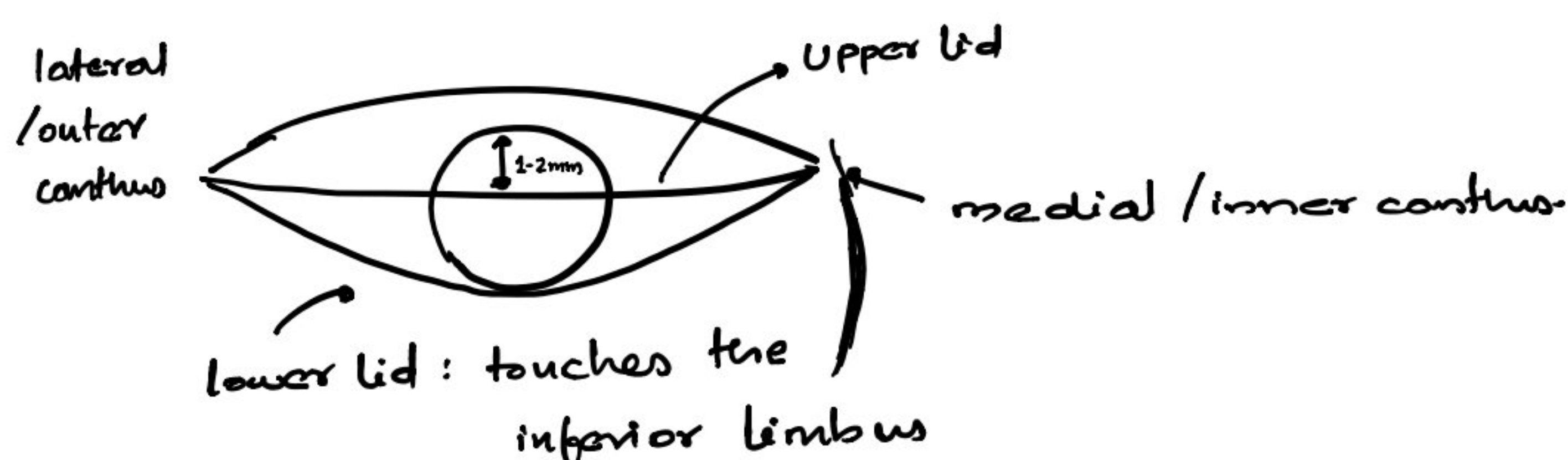
- | | | |
|---|---|---|
| 1) Social Blindness $\approx$ Blindness | → | $< 3/60$ in better eye |
| 2) Economic Blindness | → | inability to count fingers from 6m. |
| 3) Manifest Blindness | → | $1/60$ to perception of light |
| 4) Absolute Blindness | → | No perception of light |
| 5) Curable blindness | → | Blindness that can be cured
eg: Cataract |

Centers according to Vision 2020

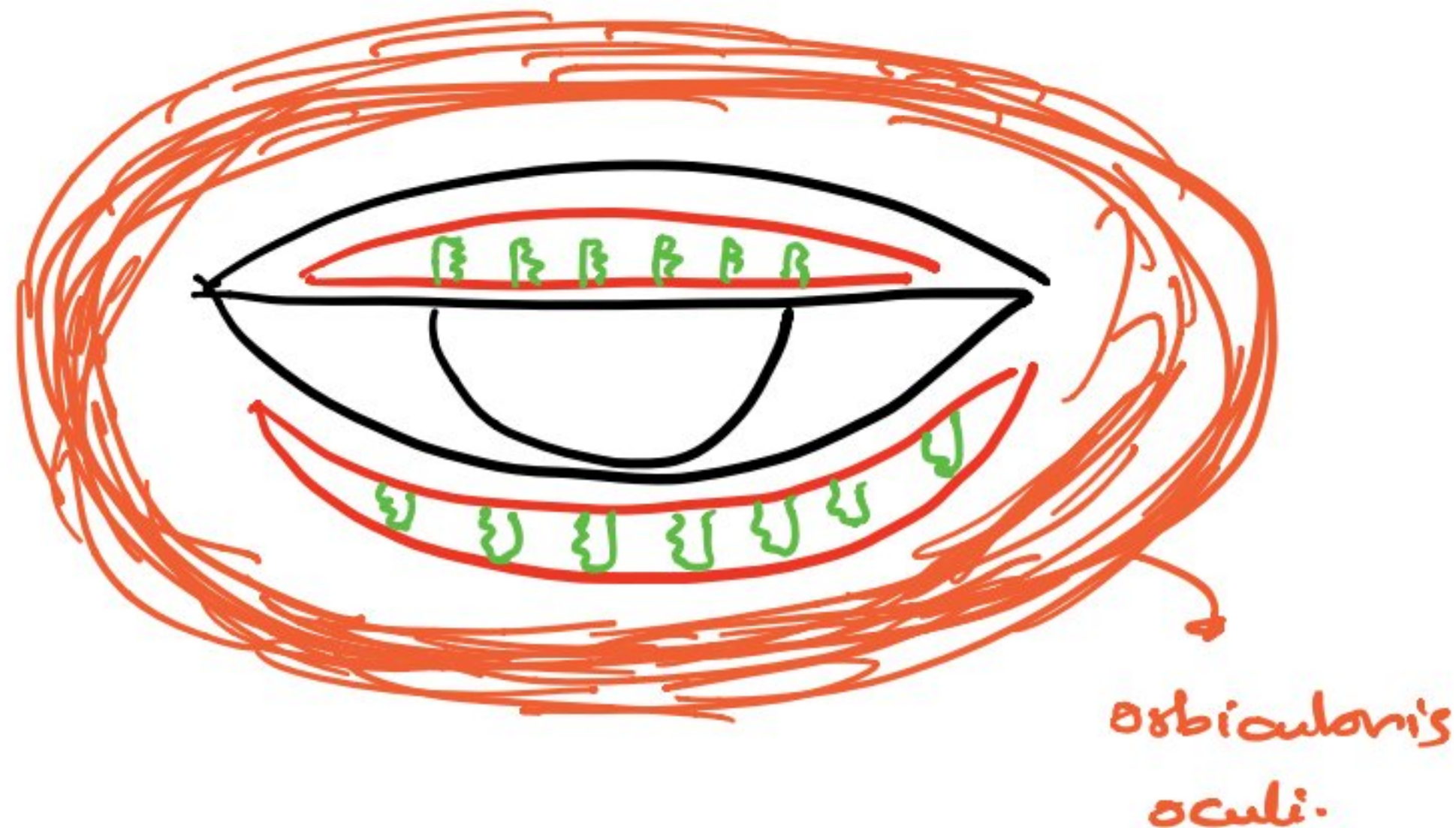
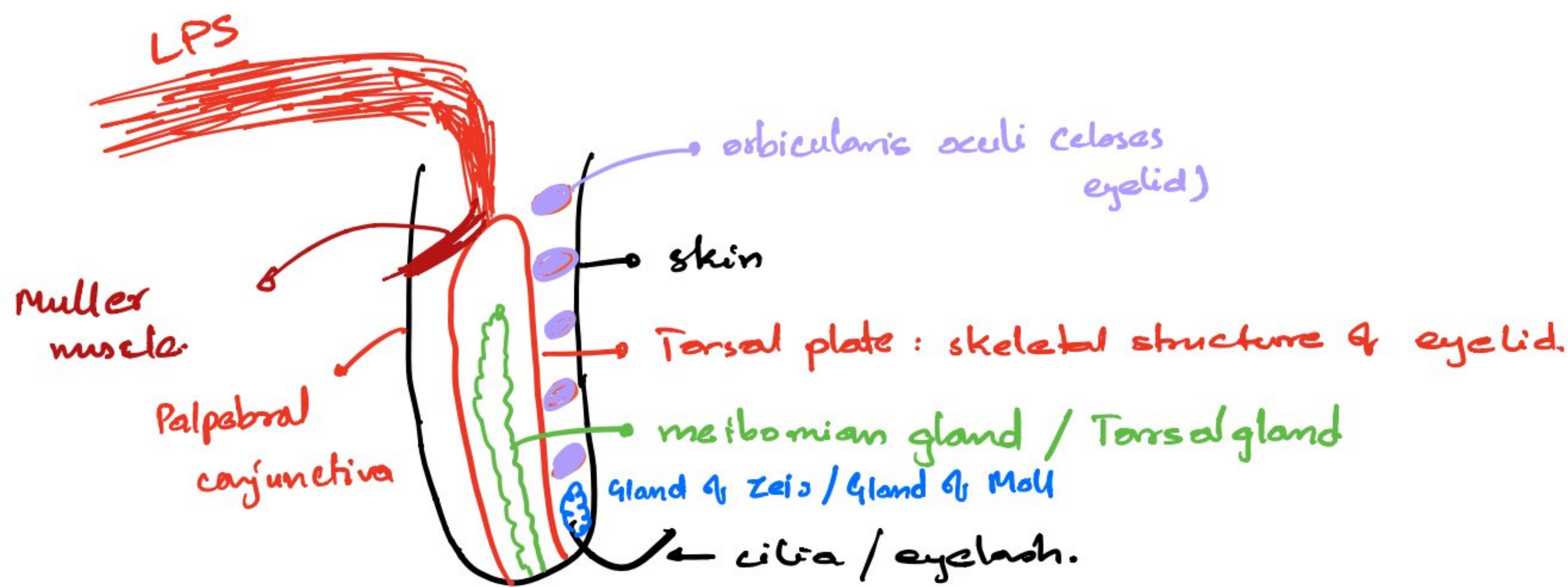
- | | | |
|-------------------------------------|---|--------|
| Centre of Excellence | → | 20 |
| Training Centre | → | 200 |
| Service Centre | → | 2000 |
| 1 st Level vision centre | → | 20,000 |

Orbicular appendages

LIDS



Cross-section of Lid:

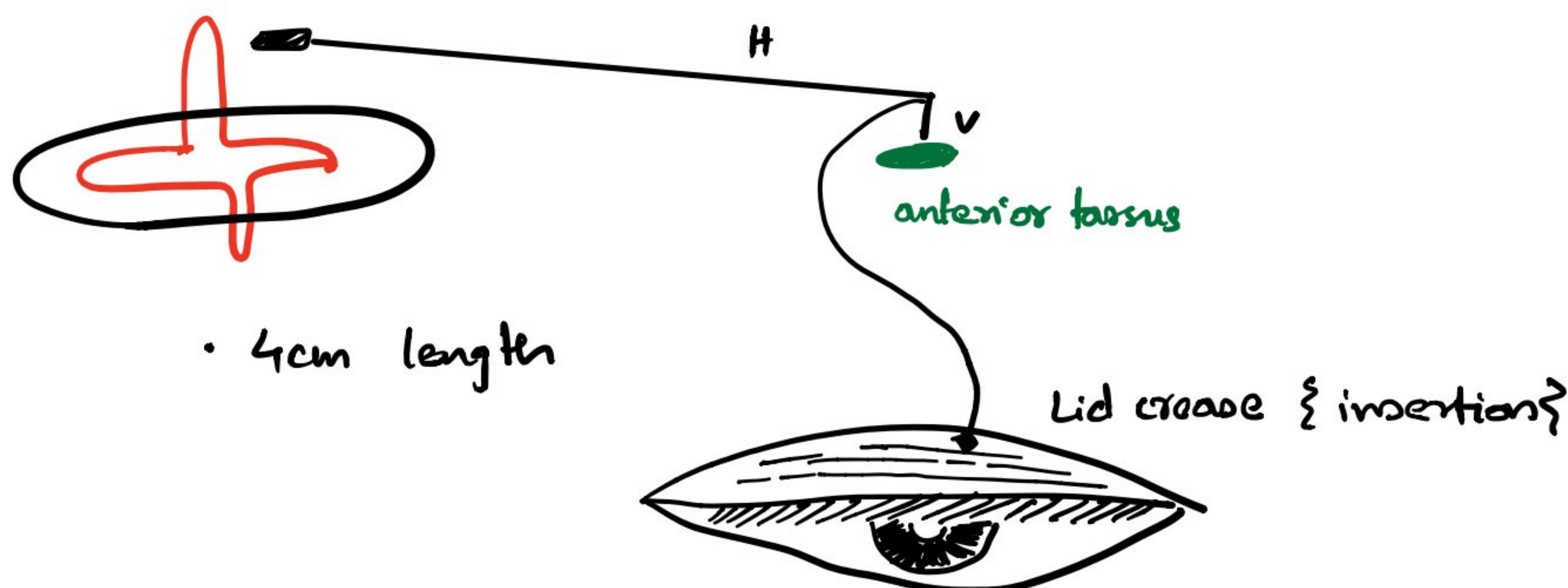


Meibomian gland → oily portion of tears (sebum)
modb. sebaceous gland

- Gland of ZEIS → modb. sebaceous gland
- Gland of MOLL → modb. apocrine sweat gland

MUSCLES:

LPS: Origin = **lesser wing of sphenoid**
 insertion = Tarsal plate (superior border)
 N.S = IIIrd N.



Accessory Muscles → Lid opening

- 1) MULLER'S muscle : N.S = sympathetic
- 2) Frontalis muscle : Facial N.

Muscle → Eyelid closure

Orbicularis Oculi ⇒ N.S : Facial N.

Inflammation of lid glands :

1) A/c suppurative inflammation ⇒ HORDEOLUM

HORDEOLUM EXTERNUM : Inflammation gland of Zeis

- Painful swelling of lid margin
- Pus pointing @ base of eyelash
- a.k.a stye → *Staphylococcal infection of hair follicle > gland of Zeis*

Rx : Epilation & Removal of eye lash



STYE

2) H. INTERNUM : Inflammation Meibomian gland

- Painful nodule away from lid margin ⇒ *On palpebral conjunctiva*



Rx : I & D vertical incision

II) C/c LIPOGRANULAMATOUS inflammation

- Chalazion : On meibomian gland
Painless nodule away from lid margin

Rx : • I & Curettage
(vertical)

- Intralesional Inj. of steroid → Triamcinolon Acetonid

Recurrent hordeola / chalazion

causes :

- i) Children : Uncorrected refractive error
- ii) Middle aged : T2 DM
- iii) Old age : Recurrent Chalazia

↓
r/o Sebaceous Gland carcinoma

HPE of curettage material

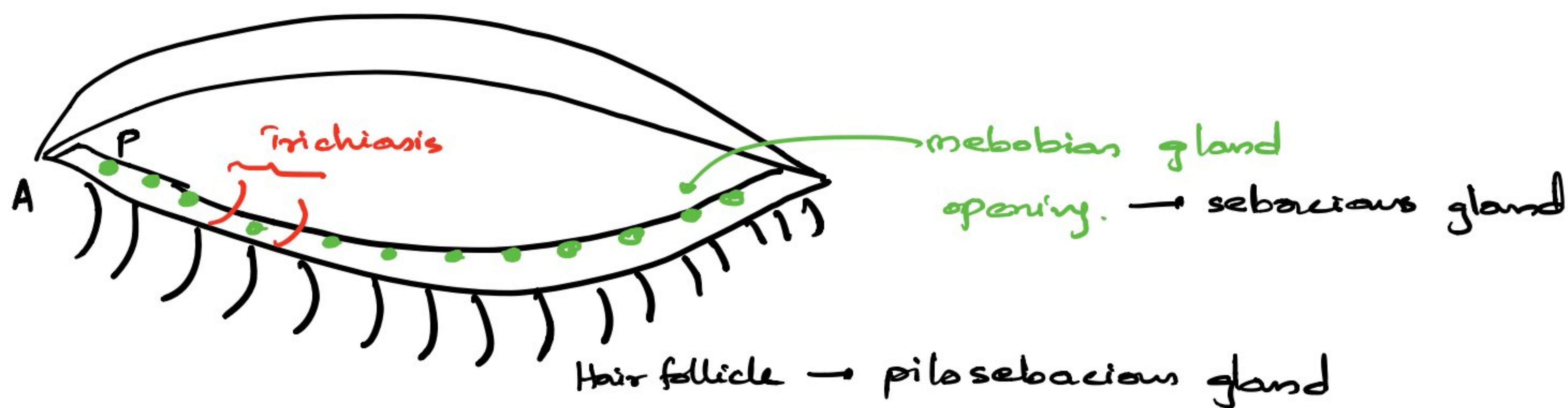


H. Internum



CHALAZION

D/o of Lid margin:



Trichiasis → Misdirected eyelash

Entropion

→ eyelid turned in ⇒ we get **pseudo trichiasis**

→ MCC is senile

Sx: Jones operation

Weiss

↓
rub on cornea

↓
Reflex tear secretion
"watering"

ENTROPION



Types:

i) Involutional entropion ⇒ Lower lid retraction, get disinserted from tarsal plate

ii) Cicatricial entropion ⇒ scar on conjunctival side } **upper eye lid**
ARTL's LINE in trachoma

Surgery for Entropion

i) Quickert suture / procedure

ii) Jones procedure

iii) Weiss procedure

Lower lid retractors → Tucking

"Quick Jones is wise entrepreneur"

Ectropion ⇒ eyelid turned out

Rx = wedge resection
medial conjunctivoplasty

Symptom: Epiphora ⇒ overflow of tears

ECTROPION



causes:

i) involutional → senility

ii) Cicatricial d/t scar

iii) Paralysis → weakness of orbicularis oculi ⇒ Bell's palsy

iv) Mechanical → d/t tumour

c/c ectropion ⇒ keratinisation of exposed palpebral conjunctiva

g: Medial conjunctivoplasty is the surgical modality of Rx used for

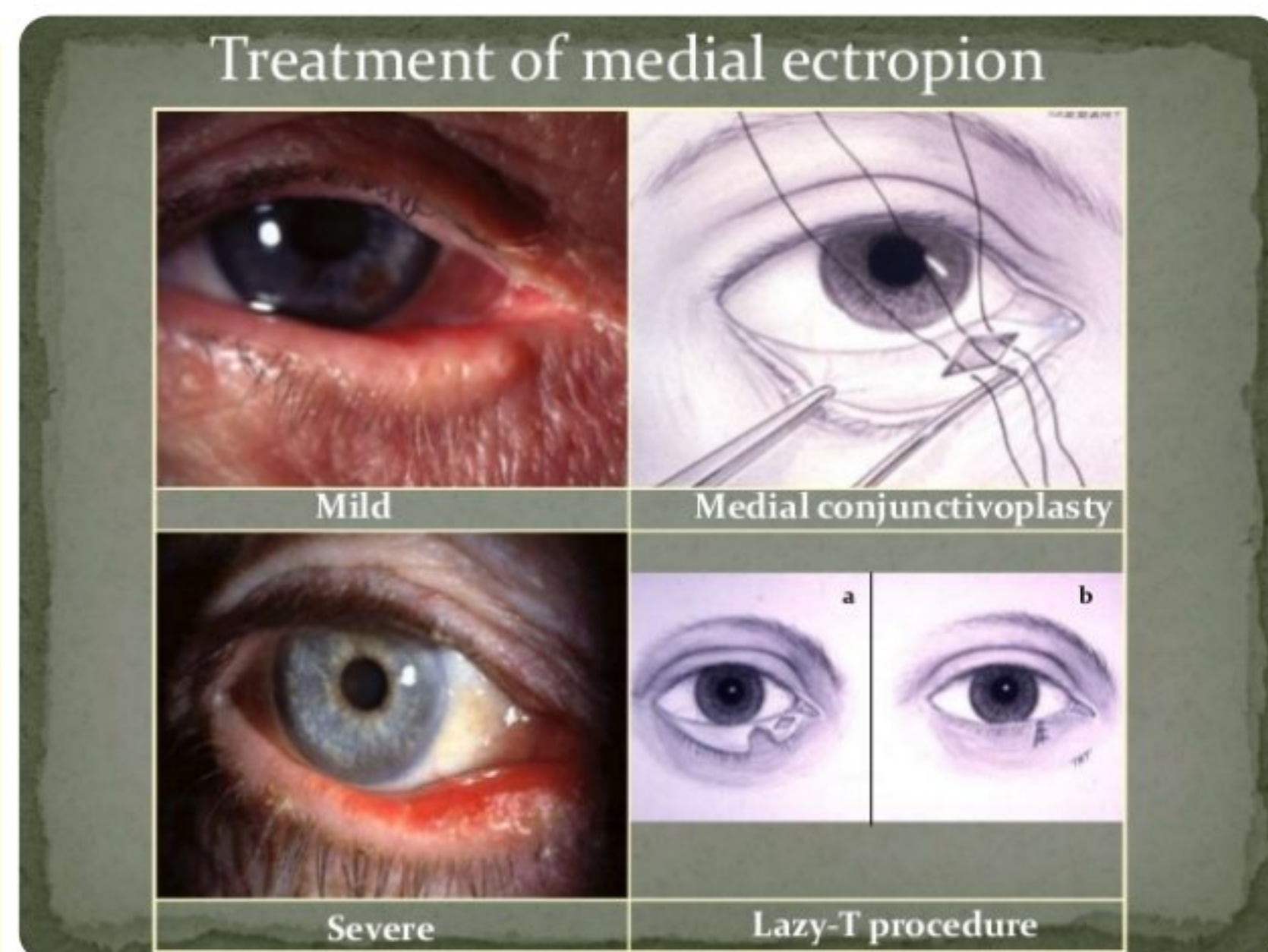
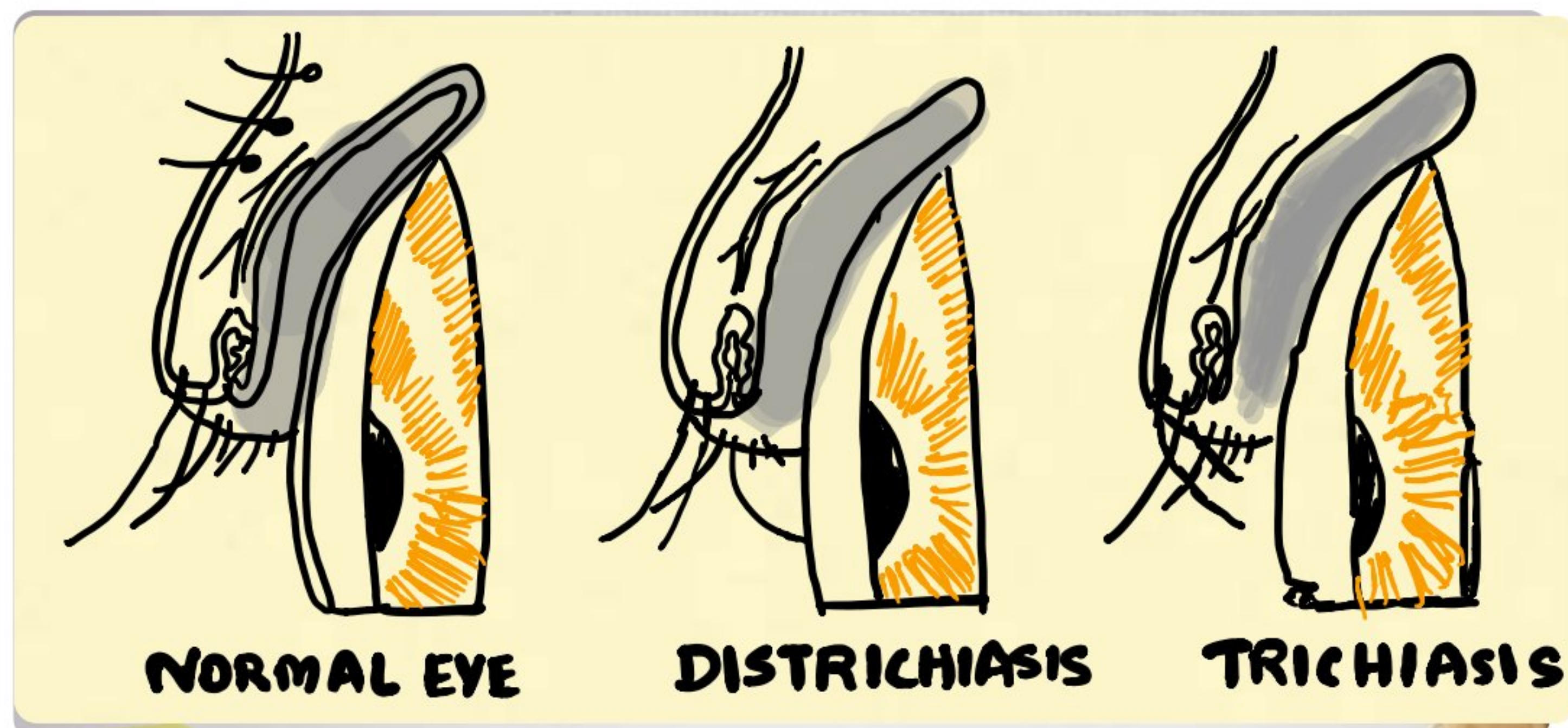
a) Entropion

b) Ectropion

c) Blepharophimosis

d) Ptosis

D/o of eyelashes



NEET '20

Distichiasis ⇒ Abn extra row of lashes

• congenital

• d/t metaplasia of meibomian gland

Trichiasis : Rx ⇒ Electrolysis of eyelash
Rx Entropion

Madarosis : loss of eyebrows / eyelashes

causes:-

1) Hypothyroidism

2) Leprosy

3) Syphilis

4) SLE

5) Trichotillomania (psychiatric)

Poliosts

→ vitiligo

→ Vogt kayangi Harra

{HLA DR4 associated panuveitis}



Hypertrichosis : ↑ sed no. of eyelashes

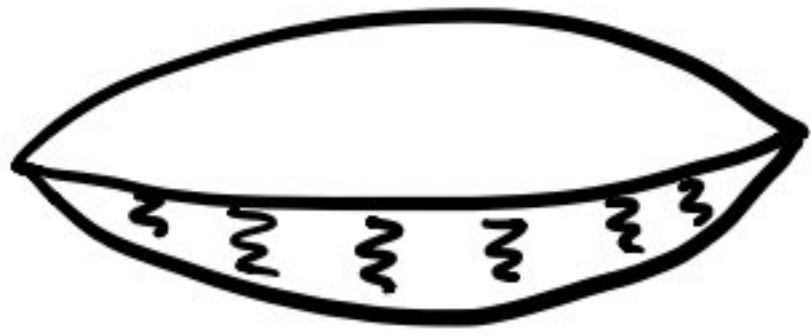
Trichomegaly : long & thick eyelashes
• S.E or PG analogue

Tylosis : • Thickening of lid margin
• Recurrent hordeolum
• Blepharitis
• Trachoma

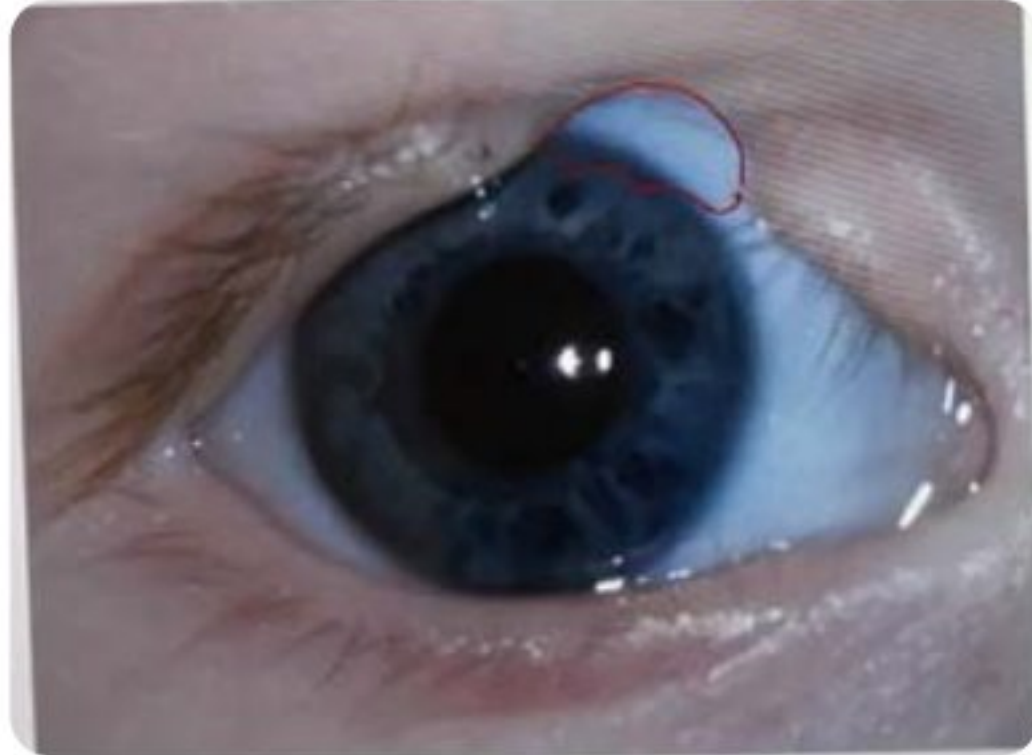
TRICHOMEGALY



Ankylo blepharous : • adhesion b/w eyelid margin
• congenital



Lid Coloboma : Tissue defect.
congenital
(MC) location → U/L



s/i TREACHER COLLIN SYNDROME → L/L coloboma

Lagophthalmos



Orbicularis Oculi weakness

← Facial Nerve Palsy

Inadequate / incomplete lid closure

when Bells phenomenon ⊖



Exposure keratitis

Rx: i) lid taping @ night
ii) lubricants
iii) Tarsorrhaphy



a) lateral T'rrhaphy
b) Median / central
c) Medial T'rrhaphy

Chemical Tarsorrhaphy

→ Botulinum Toxin A
injected on LPS
induce ptosis

LPS ⊖



ptosis

Congenital

- LPS dystrophy
- Absence of lid crease
- lid lag sign ⊕



LPS - SR & common embryological origin

LPS + SR wkness → Double elevator Palsy

↳ To be noted that

Marcus gunn jaw winking phenomenon

→ misdirection of nerve supply
Nerve to lateral pterygoid & Branch V₃

↳ LPS
midirected



DrRajaratna Thangavel

I have clarified so many times this one! It is SR plus IO. LPS may be associated bcos of common embryological origin of SR n LPS



Notespaedia.com
More than just notes

Jaw drop → eye open } SYNKINESIS.

Dx: 1) Absent lid crease (skin insertion of LPS ⊖)

LEFT UPPER LID PTOSIS



Cong ptosis

2) Lid lag sign ⊕

3) LPS fnx → v. poor

- ILIFF's method in children
- BERKE's method in adult

Myogenic Ptosis → **Myasthenia Gravis**

- AutoAb to ACh receptors @ NMJ
- U/L or B/L ptosis
- Ptosis worsen towards the end of the day
- VARIABILITY OF PTOSIS** → hallmark of M.G
- ice pack test → ptosis improves { neutralises Ab }
- Coogan's lid twitch sign → overshooting of eyelid { during attempted upgaze }

Dx: Auto Ab measurement

EMG → Repetitive stimulation of single muscle fibre → Fatigue

Lambert Eaton Sx → ⊖ EMG

Myotonic Dystrophy

- Facies :
- B/L myogen ptosis
 - B/L overarched eyebrows
 - Frontal Baldness
 - Beak like nose
 - Expression less facies
 - Christmas tree Cataract

MYOTONIC FACIES



Myotonic Dystrophy

CPEO : • c/c progressive external ophthalmoplegia

- B/L EOM + LPS ⊖
- Restricted ocular motility + B/L ptosis
- a/w **KEARS - SAYRE Syndrome**
 - CPEO
 - Cardiac conduction defect
 - Retinitis Pigmentosa

B/L ptosis:

- M.G
- Myotonic Dystrophy
- CPEO

Involucional ptosis (senile) / Aponeurotic ptosis²

- Aponeurosis of LPS disinserted from TP
- MCC of ptosis → affects lower lid
- Rx: plication of lower lid retractors

Mechanical ptosis : i) Eyelid edema
ii) Eyelid tumour a/w NF1

Management of ptosis :

- 1) Rx of M.C → No Surgery
Medical Management
- 2) Ptosis of Horner's syndrome
↳ FASANALLA - SERVAT Sx : Torso - Mullerectomy
- 3) Involucional ptosis → LPS Advancement / plication

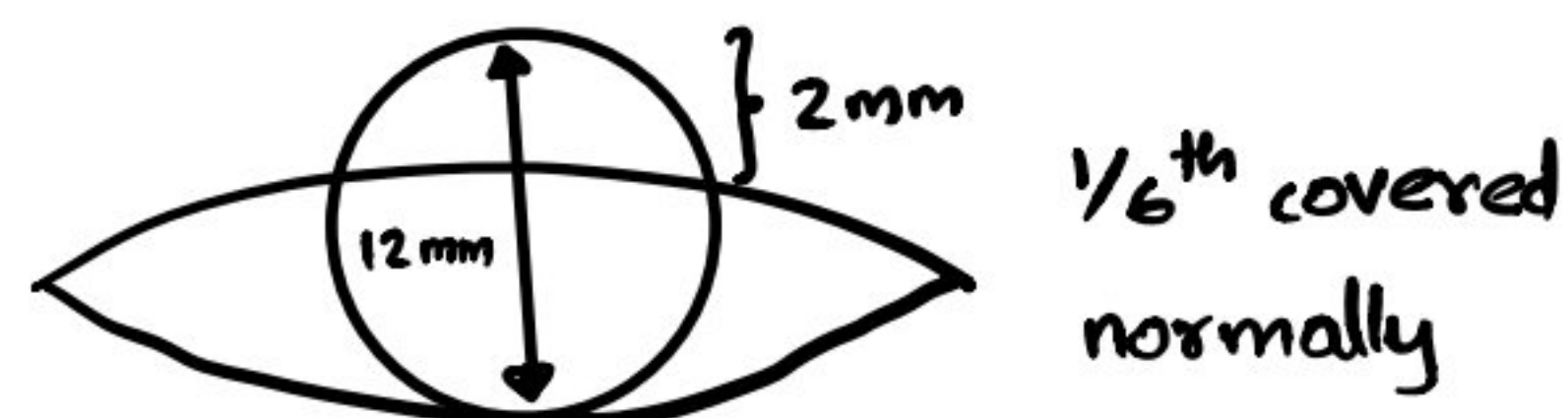
Amount of ptosis : B/L ptosis = $\frac{1}{2}$ [Distance of cornea covered by upper eyelid] - 2mm

Mild ($\leq 2\text{mm}$)
Moderate ($< 3\text{mm}$)
Severe ($\geq 4\text{mm}$)

Rx :

- LPS plication
- LPS resection
- LPS replacement

Poor LPS
action

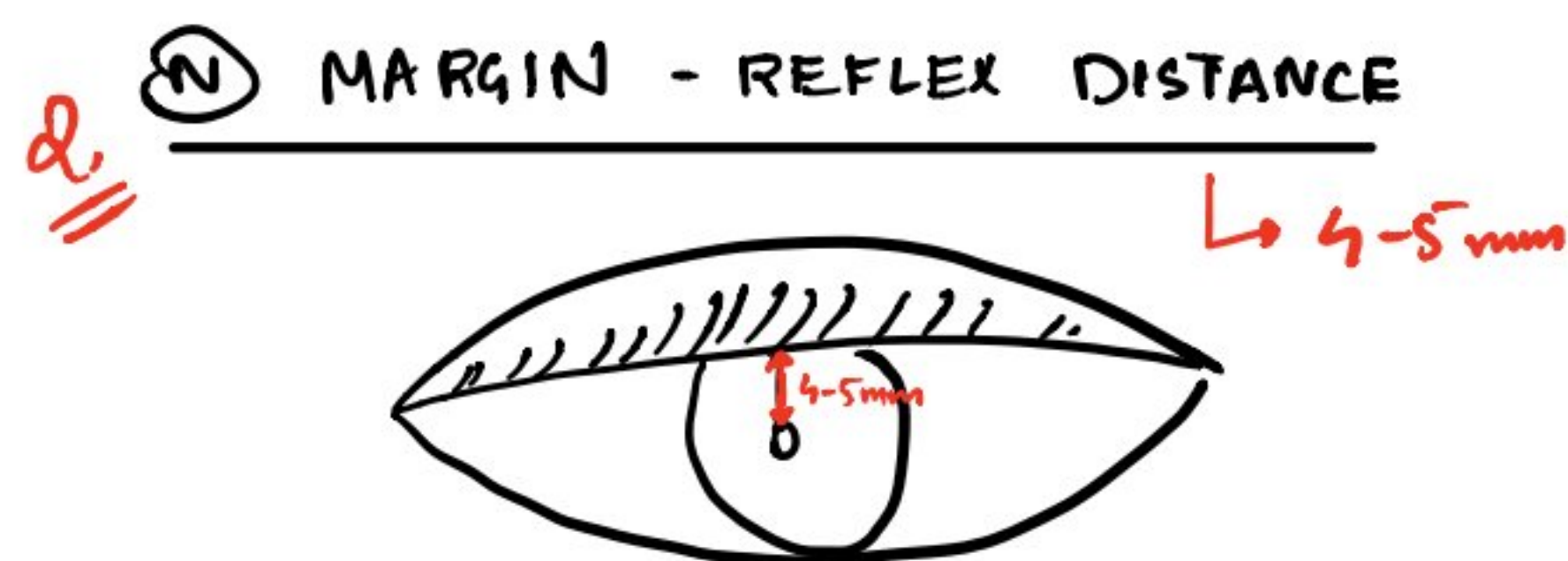
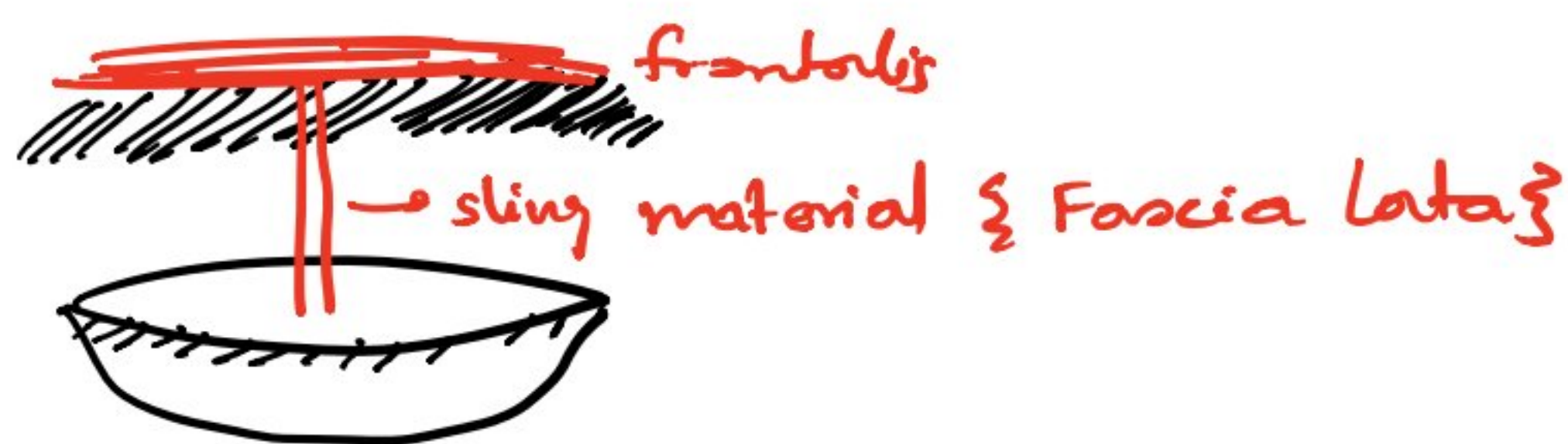


Q: Avg. height of palpebral fissure → 10mm

LPS resection

- ⇒ i) BLASKOVICS Sx = conjunctival approach
ii) EVERBUSCH Sx = skin side approach

4) Congenital ptosis = Rx : LPS replacement : Fascia lata sling operation



Q: A patient present presents with B/L ptosis. On examination, the upper 4mm of the cornea is covered by the upper eyelid. The grade of ptosis is ?

- a) Mild
- b) Moderate
- c) Severe
- d) Profound

Blepharophimosis: { Ptosis
+
↓ Horizontal PF width

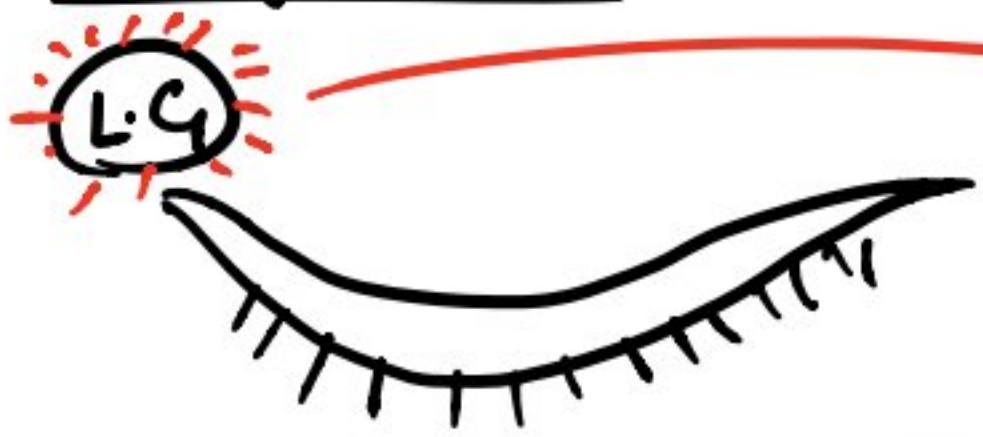
- Epicanthus inversus → from lower eyelid to up
- Telecanthus → ↑ distance b/w medial canthi.
① Interpupillary distance
 $MCD > \frac{1}{2} IPD$

- Flat nasal bridge
- Hypoplastic superior orbital rim
- Ectropion
- (AD), Familial inheritance

Hypertelorism:

- Widely spaced bony orbits
- ↑ MCD, ↑ IPD

S shaped lids



- ① Inflammation of L.G. = Dacryoadenitis
- ② Plexiform neurofibroma of U.L. ⇒ NF-1

S-shaped lid



Blepharitis: Inflammation of lid margin

↳ squamous type: Whitish flaky material @ base of eyelash - "Dandruff like"

Squamous Blepharitis



↳ Seborrhic type: Yellowish oily crust on eyelashes



Blepharitis acarica



Lid Tumours:

• (MC) lid tumour → papilloma

• (MC) lid malignancy → BCC { **VISMOTECIB** drug given for ocular BCC

Location ::

(MC) → 1) lower lid
2) Medial canthus

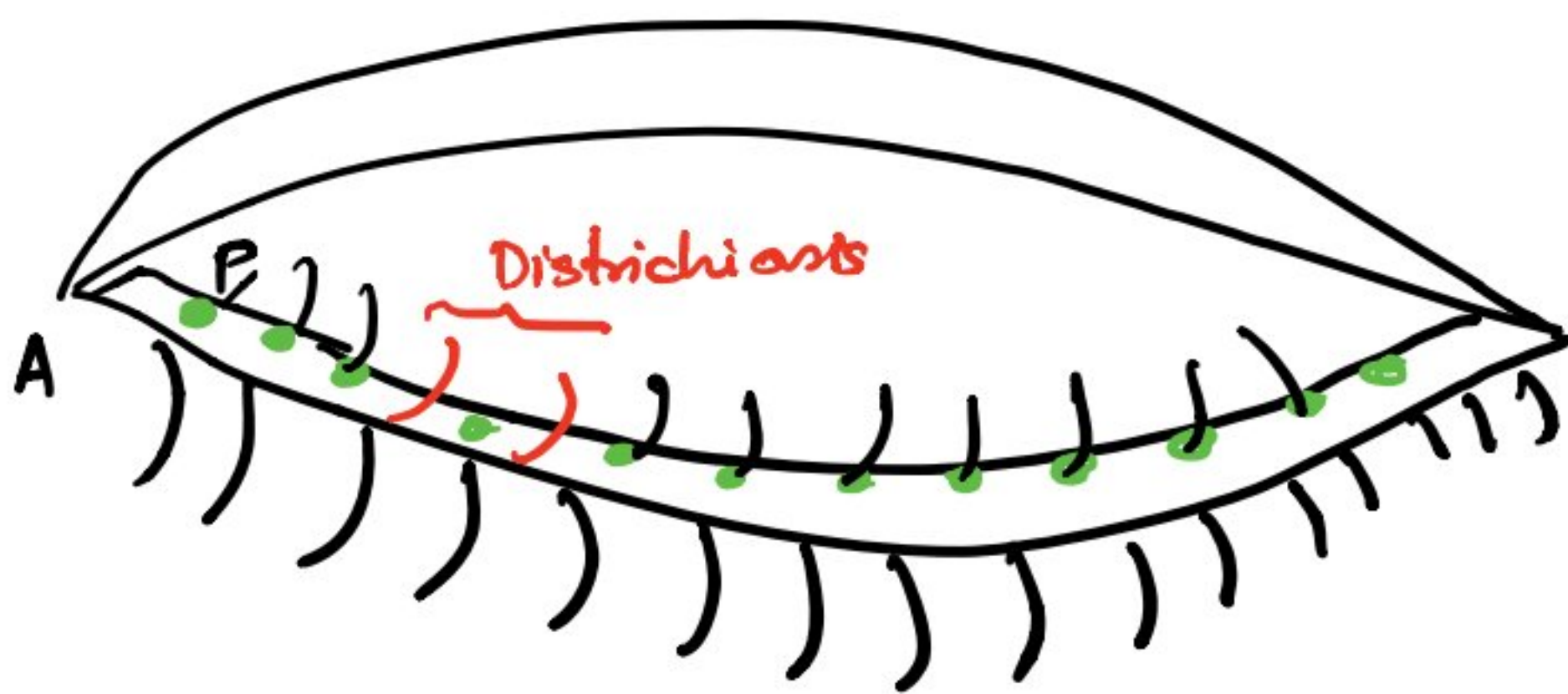
• Sebaceous gland Ca → (MC) location: Upper lid

• (MC) lid tumour in HIV patient = Kaposi's Sarcoma

Q: Blepharitis acarina is caused by
a) Streptococcus
b) Staphylococcus
c) Propionibacterium
✓ d) Demodex folliculorum

NEET '20

Distichiasis : more than 1 layer of eyelash metaplasia of meibomian gland.



Rx {
• epilation
• diathermy
• Cryotherapy.

Glands secrete Tear films (3 layers)

Lipid layers ⇒ Sebaceous gland
L Meibomian
L Zeiss
L Hair follicle

Chalazion → • Meibomian gland blockage
• upper > lower eyelid
• Painless swelling
• Lipogranulomatous inflammation

(MC) of Chalazion → eyelid rubbing d/t uncorrected Refractive error.

Rx: Intralesional steroid
or
Incision & curettage

Infected chalazion → Internal hordeolum

external hordeolum → • sty

• infection of hair follicle > Zeiss gland

Rx: oral antibiotics.

Structure of the Tear film

